High-Frequency Radio Observations of Galaxy Cluster Minihalos

by

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Abstract

Diffuse radio sources have been found within the cool cores of a growing number of relaxed galaxy clusters. These faint "minihalos" are a result of synchrotron-emitting, highly relativistic electrons. With extents over several hundred kiloparsec, they are far beyond the expected extent of a single source of relativistic electrons. The processes of *in-situ* turbulent reacceleration and hadronic production have been proposed to explain this phenomenon. Spectral steepening is expected at high radio frequencies for the reacceleration scenario, however, the majority of previous studies have been performed at lower radio frequencies (144 MHz to 1.4 GHz) where this is difficult to detect.

To investigate potential steepening, in this thesis I calibrate and image Very Large Array observations at 10 GHz of 12 clusters hosting minihalos. This represents the first steps towards a high frequency survey of minihalos. Source-subtracted measurements of minihalo flux are performed for 4 clusters, enabling comparison with prior observations at lower frequencies. Steepening was not observed for RXJ1720+2638, with a spectral index $\alpha = 1.07 \pm 0.05$, suggesting a hadronic origin for the minihalo. The remainder of the sample requires more extensive analysis.

Challenges encountered during the calibration and imaging are discussed, and recommendations for future analysis of high frequency minihalo studies are presented.

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Chapter 1

Introduction

When looking up into the sky on a clear night, those without an interest in astrophysics might imagine many of the points in the sky as galaxies just like our own, floating alone in the deep void of space. Although true to an extent, it doesn't paint an accurate picture of the cosmos. In reality, many of these are associated with galaxy clusters, massive structures of tens to thousands of individual galaxies that are gravitationally bound into a single system (somewhat analagous to the planets in our solar system, but on a vastly different scale).[1] Galaxy clusters are the largest gravitationally bound structures in the universe, with total masses of order 10^{14}–10^{15} $M_\odot$ and diameters larger than 1.5 million parsec.[2]

One might naturally expect that the majority of the cluster mass originates from the innumerable galaxies held within, however, only a few percent of the total mass comes from this baryonic matter[3] (such as stars, planets, etc). The remainder of the baryonic mass comes from a hot diffuse gas (10^7–10^8 K) spread throughout the cluster called the intracluster medium (ICM). This dilute gas ($\sim 10^3$ particles m^{-3}) is held together by the gravitational pull of the cluster, with the highest density in the core and sparsest at the edges (Meekins et al., 1971; Serlemitsos et al., 1977). Even then, this amount of visible baryonic matter only comprises 15% of the gravitational mass calculated from the dynamics of the cluster, a huge discrepancy that was a key piece of evidence for the existence of dark matter. Indeed, the rough consensus from decades of study on galaxy clusters has $\sim 3\%$ of the mass budget taken by galaxies, $\sim 12\%$ from the ICM, and a staggering $\sim 85\%$ from unobservable dark matter (Blumenthal et al., 1984; Vikhlinin et al., 2006; Pratt et al., 2019).

Between these clusters, long filaments of galaxies span across space, with their own warm-hot intergalactic medium (WHIM) mimicking the role of the ICM. The WHIM is cooler and

[1] Smaller clusters with only tens of galaxies are often known as groups.

[2] For those unaccustomed to astrophysical units, $M_\odot$ is the solar mass and defined as $1\ M_\odot \approx 10^{30}$ kg. A parsec is defined as $1\ \text{pc} \approx 3 \times 10^{16}$ m. These unusual units are far more suitable at the scale of the universe.

[3] Astronomers use the term 'baryonic' to broadly refer to all matter that is not 'dark' matter, rather than the proper particle physics definition.

more diffuse than the ICM, with densities $\leq 10^2$ particles m^{-3} and temperatures 10^5–10^7 K. These filaments are far larger than clusters, though not gravitationally bound, stretching up to hundreds of Mpc to create a cosmic web throughout the universe with galaxy clusters located at intersections between filaments (Böhringer et al., 2025). This cosmic web is the largest structure in the universe, a vast network of dark matter that we can trace via the observable galaxies and clusters (Bond et al., 1996; Pimbblet, 2005) with ongoing work in detecting and understanding these filaments observationally (Tudorache et al., 2025). Clusters are known to form in part due to accretion of the WHIM at intersections of the cosmic web, as well as by mergers between smaller clusters (Press and Schechter, 1974; Voit, 2005; Kravtsov and Borgani, 2012).

An important distinction should be made when analysing clusters, between "disturbed" clusters undergoing merger events or still in a turbulent state from a recent merger, and "relaxed" clusters which have settled into an equilibrium state. Relaxed clusters favour theoretical studies due to their greater spherical symmetries, fewer substructures, and therefore, simpler modelling (Mantz et al., 2016). Many of these relaxed clusters also feature a "cool core", where the dense central region of the cluster exhibits a lower temperature than the outer regions. The high density of the ICM in the core causes a large outflux of X-ray emission. The resulting cooling time may then be lower than the expansion rate of the universe (known as Hubble time), causing the core temperature to drop and an inwards flow of gas to develop (Peterson and Fabian, 2006; Hudson et al., 2010). Although cool cores are observed, there is little evidence for the predicted inward cooling flow, suggesting the existence of a counteracting heating mechanism. The most widely supported theory involves active galactic nuclei (AGN), supermassive black holes at the centre of some galaxies. AGN consume massive amounts of matter and produce radio synchrotron emission that forms jets and lobes, which have been observed interacting with the ICM to distribute energy in "radio-mode" feedback (Fabian, 2012). The exact method of this heating is still disputed and undergoing research.

Within the high mass subset of relaxed cool core clusters ($M_{500} > 6\times10^{14}$ M$_\odot$)[4] recent works have shown that 80% contain radio minihalos (Giacintucci et al., 2017). Less common in lower mass clusters, and unobserved within non-cool core clusters, minihalos are diffuse radio sources with currently unknown origins. This radio emission is localised within the cool core, with radii from 50 kpc to $0.2R_{500}$.[5] This emission cannot be directly formed by radio sources such as AGN, as the spatial extent of these halos is well beyond the expected extent for particles sourced from an AGN (e.g. Bagchi et al., 2002; Giacintucci et al., 2014b). Understanding the origins and physics behind minihalos is an ongoing process, with viable production methods proposed, but a lack of conclusive observational evidence to differentiate between them. This thesis will provide

[4]The value M_{500} refers to the total mass within a radius of R_{500}, defined by the distance at which the clusters average enclosed density is 500 times the critical density of the universe at the cluster redshift. The critical density is a cosmological quantity that has varied over the lifetime of the universe.

[5]The radius $0.2R_{500}$ is a standard metric to separate the core and the outer ICM. Within this radius, the cooling, AGN feedback, and similar processes become significant to the behaviour of the ICM.

new observational data and a generalised process to provide further evidence to differentiate between the proposed processes.

1.1 Observations of Galaxy Clusters

Different processes within the cluster, and more specifically ICM, provide information about the physical environment. This radiation is emitted across the electromagnetic spectrum and we design telescopes to capture and relay different bands in order to identify different components of what we observe. The question then becomes, how do observations at different frequency bands inform us to the structure and processes occuring within a cluster?

1.1.1 Optical

The vast majority of optical light from clusters comes from stars within the component galaxies. Although incredibly informing about the stars and galaxies themselves, in the context of minihalos, optical light has limited use. Optical observations are useful in determining the component galaxies of a cluster, alongside investigating the radial velocities of galaxies. Since the radial velocity is dependent on the gravitational potential of the cluster, the distribution allows us to estimate the clusters mass (Voit, 2005). This analysis is difficult for more distant clusters and disturbed clusters, as the dynamical state will affect the radial velocities and skew the resulting mass.

1.1.2 X-ray

The hot gas of the intracluster medium forms an optically thin plasma, which has a degree of ionisation that depends on the temperature and density. We observe an X-ray spectrum from this gas with both continuum and line characteristics, and photon energies between 0.1 keV and 100 keV.[6] The energy of the photons is dependent on the temperature of the ICM, resulting in most emissions being in the narrower range of 1 to 10 keV (Schneider, 2014).

Line emission occurs due to electron transitions between orbitals within an atom. The temperature and density of the ICM changes the measured rate of this line emission, however, this becomes limited at higher temperatures as the ionisation fraction increases. The distribution of line emissions can also be an indicator of abundance of elements such as C, N, Ne, Mg, Si, S, Ar, Ca, and Fe. This is particularly evident at lower temperatures ($\lesssim 2$ keV) where the total emissions can be dominated by line emission (Voit, 2005; Schneider, 2014).

Continuum radiation is due to brehmsstrahlung, or free-free radiation, which is where electrons are deflected via Coulomb interaction with nearby nuclei. This deflection is an

[6] X-ray emission is historically measured in kilo-electron volts (energy) rather than frequency or wavelength.

acceleration which induces a change in kinetic energy that is emitted as a photon. The spectral energy distribution of these emissions is related to the kinetic energy of the particles, and therefore, the temperature T of the ICM. The scale of emissions is similarly impacted by the electron number density n_e, which traces the gas density of the ICM. This gives rise to the following relationship to total continuum emissivity ϵ_c (Schneider, 2014).

$$\epsilon_c \propto T^{\frac{1}{2}} n_e^2 \tag{1.1}$$

X-ray spectroscopy is a powerful tool to characterise the ICM and a clusters dynamical state. In general, telescopes trade off between high spatial or spectral resolutions, i.e. between detecting exactly where a photon originated from or an accurate measurement of the photon energy. This becomes somewhat of a constraint when observing galaxy clusters as we require reasonable spatial and spectral resolutions for these large objects. One common instrument is the space telescope *Chandra X-ray Observatory* launched in 1999 . Many studies of minihalos use *Chandra* observations to map the structure of the cool core and investigate potential relationships to the radio emission (e.g. Giacintucci et al., 2024). However, due to the aging detectors and contaminant build up on *Chandra*, the sensitivity of the telescope is decreasing. This is particularly bad at low energies (< 2 keV), where sensitivity is below 10% of where it was two decades ago (Grant et al., 2024). A new telescope *XRISM* was launched in late 2023 and was designed as a successor to *Chandra*. Over the last year researchers have begun to utilise the greater resolution to more deeply probe the dynamics of the ICM in some clusters (Sarkar et al., 2025; XRISM Collaboration et al., 2025). Although not the focus of this thesis, X-ray observations provide key ancillary data in investigating the processes involved.

Cite/footnote telescopes

1.1.3 SZ

Actually write this you idiot

1.1.4 Radio

Radio observations cover several orders of magnitude in wavelength, with frequencies from 10 MHz to over 100 GHz. This massive range comes with several difficulties for astronomers, with the primary one being the angular resolution of the telescope. Angular resolution θ defines the smallest area that can be resolved given an aperture and the observing wavelength.[7] Equation 1.2 gives the formula for angular resolution in arcseconds, or $\frac{1}{3600}$th of a degree, for an aperture of diameter D and observing wavelength λ.[8]

$$\theta = 2.5 \times 10^5 \frac{\lambda}{D} \tag{1.2}$$

Radio waves range from 3 cm to 3 km, orders of magnitude larger than optical ($\sim 1\mu$m) and X-ray (0.01–10 nm). It is then clear that to resolve the same object at radio frequencies as X-ray

[7] The angular resolution of a human eye (with 20/20 vision) is 1 arcminute. *See note 7*

[8] Arcseconds (”), arcminutes ($1' = 60$”), and degrees are common units in astrophysics as most angles are measured with this unit.

or optical, vastly larger telescopes would be necessary. In order to observe over this range several types of telescopes have been developed, including single dish telescopes, interferometer arrays, and very long baseline interferometer arrays (VLBI).

The size of a single dish antenna is limited by both practical construction and from scientific usability. The largest dish is the *Five-hundred-meter Aperture Spherical Telescope* (FAST) located in Guizhou China, though of the 500 metre diameter only a 300 metre area is usable at a time. Furthermore, as an immovable dish, it is limited in its view of the sky, with a maximum angle of 40 degrees from zenith. Several smaller 100 metre diameter dishes exist such as the *Green Bank Telescope* in West Virginia United States, and the *Effelsburg Telescope* near Bonn, Germany. These dishes also have a limited view of the sky and variable amounts of steerability due to their size, alongside worsening resolution at increasing frequencies. These restrictions developed into interferometry, the use of an array of single dish telescopes in tandem to simulate a larger dish. The *Karl G. Jansky Very Large Array* (VLA) in New Mexico United States, is comprised of 28 fully steerable 25-metre radio telescopes mounted on railway tracks in a Y-shaped array, observing between 73 MHz and 50 GHz. The *MeerKAT* telescope in the Meerkat National Park of South Australia contains 64 13.5-metre telescopes and operates between 1000 MHz and 10 GHz. These arrays can simulate observations by dishes several kilometres in diameter, while simultaneously able to track objects across the entire observable sky. VLBI takes this concept even further by using radio telescopes across different continents, synchronised by atomic clocks, to simulate dishes comparable to the size of the Earth. This thesis uses observations taken by the VLA, and the physics of these telescopes will be discussed in more detail in Chapter 2.

Cite/footnote telescopes

Radio emission detected by these telescopes from clusters falls into the category of 'synchrotron' emission. This synchrotron emission is produced by particles travelling at relativistic velocities through a magnetic field, emitting photons at radio energies characterised by their velocity and the magnetic field strength (Feretti et al., 2012; Schneider, 2014). This is due to the Lorentz force ($\boldsymbol{F} = \boldsymbol{v} \times \boldsymbol{B}$) perpendicular to their direction of motion, which induces a helical path through the field, and therefore constant acceleration. This is analagous to brehmsstrahlung radiation, but occurs due to a magnetic interaction instead of electric, which is why it is somewhat rarely called 'magnetobrehmsstralung'. A more detailed discussion of the types of radio emission is given in Section 1.2. A review of the physical processes that give rise to this emission, and the processes spectral and spatial markers is given in Chapter 2.

1.1.5 Gamma

The last region of interest for minihalo research is in detecting gamma radiation, or gamma-rays. This is the high energy limit for astronomy, where telescopes are designed to detect photons with

energies in the MeV and GeV range.[9] These prove difficult to not only detect, due to the majority of rays being absorbed in the atmosphere, but also to accurately describe their origin (Schneider, 2014). The *Fermi Gamma-ray Space Telescope* was built specifically for this research and was launched in 2008 carrying the Large Area Telescope (LAT), with an energy range between 30 MeV and 300 GeV.

There are various sources for gamma-rays, such as Gamma Ray Bursts (GRB), which are extremely short and intense flashes. Many of these have been spatially located to coincide with bright AGNS called blazars, and rapidly spinning magnetised neutron stars called pulsars. Another source is from gamma-ray background radiation, which is expected via neutral pion decays resulting from proton-proton collisions. Attempts to detect gamma-ray emissions connected to a clusters ICM have been ongoing as they allow us to put limits on effectiveness of proposed mechanisms to create radio halos (e.g. Reimer et al., 2003; Ackermann et al., 2010; Zandanel and Ando, 2014). Though progress has been made for giant radio halos, in particular firm constraints for the Coma cluster (Brunetti et al., 2012), minihalos have a smaller spatial extent and are much fainter, with no conclusive studies. The role of gamma rays in determining the processes creating minihalos is detailed in Chapter 2.

1.2 Diffuse Radio Emission

It has long been known that some clusters exhibit diffuse radio synchrotron emission that cannot be directly associated with radio galaxies within the cluster. This synchrotron emission can take many spatial arrangements, with the primary categories being radio halos, radio relics, and revived AGN sources. Radio halos are extended sources that trace the ICM distribution and operate at the scales of giant radio halos, matching the full ICM extent, and minihalos that are contained within a clusters cool core. Relics instead reflect particles that have been reaccelerated to relativistic energies by shock waves travelling through the ICM, such as the violent energies released during merger events. Relics tend to be elongated and irregularly shaped, located at the clusters periphery instead of the centre. Revived AGN sources are radio plasma that has been emitted from an AGN and re-energised similarly to the radio relics. From this point, this thesis will focus on radio emission in the context of halos, and in particular, minihalos. Several types of diffuse radio emission are shown in Figure 1.1.

1.2.1 Production

A key question in relation to the radio halos is how they are produced. The relativistic particles producing this emission have a characteristic lifetime due to the energy losses as they travel through space, and therefore a limited diffusion length-scale in which we should be able to

[9] Like X-ray, gamma-rays are measured by their energy rather than frequency.

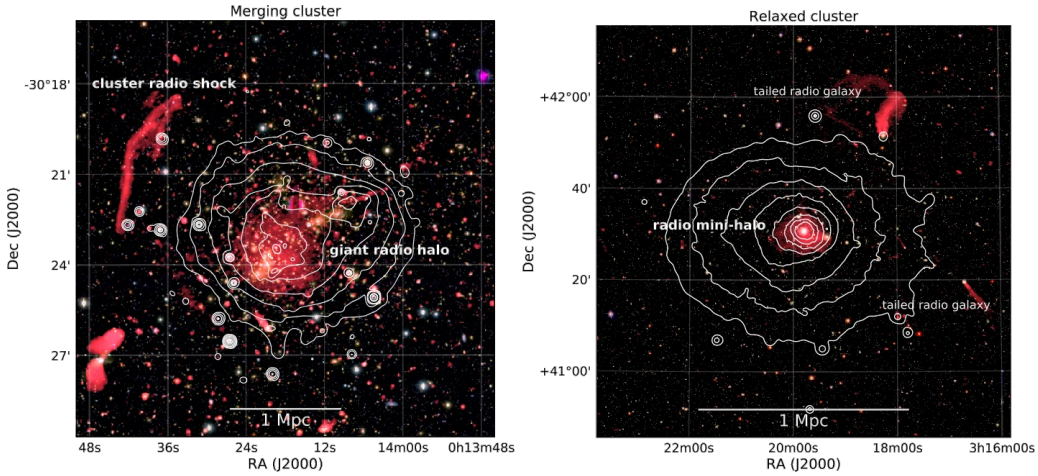

Figure 1.1: Left panel: Merging galaxy cluster A2744, VLA 1-4 GHz image (red) and Chandra X-ray contours (white) overlaid on Subaru BRz optical image. X-ray surface brightness contour are drawn proportional to $[1, 4, 16, 64, \ldots]$. The luminous giant radio halo and a cluster radio shock (relic) are labelled. Right panel: Relaxed cool core Perseus cluster, VLA 230-470 MHz image (red) and XMM-Newton X-ray contours (white) in the 0.4-1.3 keV band are overlaid on an optical image. X-ray contours have same spacing as the left image. The Perseus cluster hosts a radio mini-halo as well as two head-tail radio galaxies. (Figure 3 of (van Weeren et al., 2019))

observe them. Within a weak magnetic field ($\sim 10^{-7}$ G) these distances are about 10 pc, which even accounting for bulk motion of the ICM and more favourable diffusion environments are well below the Mpc scales observed in many clusters (Bagchi et al., 2002). Two primary theories have been brought forward to explain this phenomenon, one being turbulence in the ICM causing "reacceleration" of once relativistic particles (e.g. Brunetti et al., 2001; Gitti et al., 2002; ZuHone et al., 2013; Zandanel et al., 2014). The other is inelastic collisions between relativistic cosmic ray protons and thermal protons, termed "hadronic" or secondary production (e.g. Dennison, 1980; Dolag and Enßlin, 2000; Keshet and Loeb, 2010; ZuHone et al., 2015; Ignesti et al., 2020).

Both of these mechanisms produce similar observable features, which makes disentangling the true source much more challenging. Investigation into spatial extents and spectral features of giant radio halos has given the reacceleration theory more credence than hadronic production (Brunetti et al., 2012), however, the debate is still ongoing for minihalos. A detailed explanation of these processes is provided in Section 2.3. Due to their faint nature and smaller size, observations require high sensitivity and resolution in order to accurately capture an image.

1.2.2 Minihalo detections

Much effort has been devoted towards detection and validation of minihalos in order to build a comprehensive sample. Over the last two decades the number of detected minihalos has increased substantially, from 10 minihalos and minihalo candidates in 2011 (Feretti et al., 2012), to 22 in 2017 (Giacintucci et al., 2014b), and 40 in 2024 (Giacintucci et al., 2024).

One of the major issues with detection is in separating the minihalo emission from the much brighter sources embedded within and nearby it. Alongside the minihalo in the core is the brightest cluster galaxy (BCG), generally the most radio-loud AGN within the cluster. Minihalos typically have a surface brightness of $1 \sim 5$ μJy arcsec^{-2}, while the BCG can have a flux 1000 times higher.[10] The BCG must then be masked or subtracted away from an observation to reveal the minihalo, but this requires the observation to have good resolution of point sources as well as the larger spatial minihalos. This requirement makes current general sky surveys insufficient for detecting minihalos, with some studies suggesting that next generation radio telescopes could reveal thousands more minihalos due to improved capabilities (Gitti et al., 2018).

1.3 This Thesis

1.3.1 Aims

The primary goal of this thesis is to calibrate, image, and analyse new radio observations of 12 minihalos at 10 GHz. These observations were performed by the VLA in C and D configuration over two periods in early 2024 and early 2025. This data constitutes the first broad survey of minihalos at high radio frequencies.

This data will complement previous surveys at lower frequencies taken by the *Low-Frequency Array* (LOFAR) at 140 MHz and MeerKAT at 1.3 GHz (Riseley et al., 2022). Processing this data will allow measurements of the minihalo flux to investigate if steepening of the spectrum occurs at higher frequency, which will assist in differentiating between the reacceleration and hadronic scenarios.

1.3.2 Work

In pursuing these aims, I downloaded the raw interferometric data from the National Radio Astronomy Observatory (NRAO) Archive[11] and documented the calibration and imaging process. I then performed various analyses on the final images for each minihalo to evaluate the limits and usefulness of 10 GHz observations for characterising the production mechanisms of minihalos.

This work was done in a source controlled repository[12] so that the results are reproducible and documented. Due to the size of the datasets, (uncalibrated $\sim$ 1.4 TB compressed, calibrated

Add final size $\sim$ 2.5 TB, imaging $\sim$? TB), this is stored on the Storage fOr Learning And Research (SOLAR) large scale storage provided by Victoria University of Wellington.

[10] The Jansky (Jy) is a unit measurement of spectral flux density common in radio astronomy. It measures the rate of energy received per square meter per hertz, or 10^{-26}W m^{-2} Hz^{-1} in SI units.

[11] https://data.nrao.edu/portal/

[12] https://github.com/MoonlitMoo/Masters-2025

The repository is split into two primary sections, *Data Processing* and *Image Processing*. The first section follows the calibration of the raw data and separation from observation runs into cluster specific datasets, split by observation id. The following image processing includes the cleaning and creation of the images from the trimmed datasets and the associated analysis and is split by cluster.

1.3.3 Cosmology

Several physical values depend on the constants chosen for the cosmological model of the universe. This work assumes a flat ΛCDM model with Hubble's constant $H_0 = 70$ km s^{-1} Mpc^{-1}, fractional density of dark energy $\Omega_\Lambda = 0.7$, and fractional density of matter $\Omega_m = 0.3$.

1.3.4 Layout

The introduction has given a brief overview on several concepts necessary for understanding what minihalos are. The remainder of the thesis is organised as follows.

Chapter 2 provides deeper explanation of synchrotron radiation, the hadronic and turbulent reacceleration processes, and a summary of previous research into minihalos.

Chapter 3 introduces the collected data and explains the calibration and imaging process used.

Chapter 4 explains the tools and techniques used in the analysis of the final images.

Chapter 5 details the results found for each cluster, with the context of previous studies.

Chapter 6

Chapter 7

Chapter 2

Radio Halos

2.1 Synchrotron emission

Since the first discoveries of diffuse radio emission, much research has been devoted towards understanding the mechanisms behind the production of synchrotron-emitting electrons. These electrons are also known as 'cosmic ray' electrons (CRe) or non-thermal electrons to differentiate them from the thermal electrons of the ICM. The CRe have an individual energy $E = \gamma m_e c^2$ dependent on their Lorentz factor γ.

$$\gamma = \left(1 - \frac{v^2}{c^2}\right)^{-\frac{1}{2}} \tag{2.1}$$

In the case of radio halos we can expect $\gamma \gtrsim 10^4$ which corresponds to velocities within 99.9999% the speed of light. Throughout their radiative lifetime CRe produce radio synchrotron emission with power dependent on the magnetic field B and their energy E. This is broadly a continuum radiation with the synchrotron power loss described by the following equation, where σ_T is the Tompson cross-section, and $U_B = B^2/(8\pi)$ is the energy density of the magnetic (Sarazin, 1999).

$$\left[-\frac{dE}{dt}\right]_{syn} = \frac{4}{3}\frac{\sigma_T}{m_e c}\gamma^2 U_B \tag{2.2}$$

They also interact with thermal protons in the ICM via inverse Compton (IC) scattering,[1] of which the dominant proton population originates from the Cosmic Microwave Background (CMB). The scattered photon has a boosted energy that is proportional to its original energy. In units of frequency this relation is:

$$\nu_p = \frac{3}{4}\gamma^2 \nu_{p,0} \tag{2.3}$$

For the CRe under consideration this results in a boosting factor $\sim 10^6$. For thermal photons sourced from the CMB, they move through the cluster with a temperature of T = 2.73 K, to

[1] Standard Compton scattering has ultrarelativistic photons interacting with thermal electrons, the opposite to what we are considering.

which the Planck function peaks at $\nu \approx 1.6 \times 10^{11}$ Hz. This interaction causes emission at $\nu \approx 10^{17}$ Hz or $\lambda \approx 1$ nm, corresponding to high energy X-rays (Sarazin, 1999; Govoni and Feretti, 2004). The energy losses due to IC scattering with CMB photons can be expressed as the CMB energy density $U_{CMB} = B_{CMB}^2/(8\pi)$, and an equivalent magnetic field strength $B_{CMB} = 3.25(1+z)^2$ [μG]. This magnetic field isn't real, but results in equivalent energy losses as if electrons were interacting with a real magnetic field within the cluster.

$$\left[-\frac{dE}{dt}\right]_{IC} = \frac{4}{3}\frac{\sigma_T}{m_e c}\gamma^2 U_{CMB} \tag{2.4}$$

These mechanisms are then highly dominated by the synchrotron and IC losses, which scale $\propto \gamma^2$, at high Lorentz factors and so can be neglected. The combined synchrotron and IC losses can then be written as a single loss equation where the magnetic fields are measured in μG. Further losses due to Coulomb and brehmsstrahlung interactions are possible, but scale with γ as follows (Sarazin, 1999).

$$\left[-\frac{dE}{dt}\right]_{Coul} \propto \ln\gamma, \qquad \left[-\frac{dE}{dt}\right]_{brehm} \propto \gamma \ln(\gamma)$$

These mechanisms are naturally highly dominated by the synchrotron and IC losses, which scale $\propto \gamma^2$, at high Lorentz factors and so can be neglected. The combined synchrotron and IC losses can then be written as a single loss equation where the magnetic fields are measured in μG.

$$-\frac{dE}{dt} = \frac{1}{6\pi}\frac{\sigma_T}{m_e c}\gamma^2\left(B^2 + B_{CMB}^2\right) \tag{2.5}$$

The standard assumptions regarding the population of CRe within the ICM are that it is homogeneous, isotropic, and follows a power-law energy distribution $n(E) = n_0 E^{-\delta}$. The total synchrotron emissivity ϵ_s at a given frequency ν is dependent on the CRe number density, the strength of the ICM magnetic field, and the spectral index of the CRe distribution.

$$\epsilon_s(\nu) \propto n_0 B^{(\delta+1)/2}\nu^{-(\delta-1)/2} \tag{2.6}$$

This can be rewritten in terms of $\alpha = (\delta - 1)/2$, the spectral index of the synchrotron emission.

$$\epsilon_s(\nu) \propto n_0 B^{\alpha+1}\nu^{-\alpha} \tag{2.7}$$

This relation shows a degeneracy between the magnetic field strength and the CRe number density, as such, the synchrotron radiation alone is insufficient to directly measure the magnetic field. Estimates of the magnetic field are generally found by minimising the total energy contributed by the relativistic electrons $(1+k)U_e$, where k is the ratio of energy of relativistic protons to the electrons, and the magnetic field U_B.

$$U_{tot} = (1+k)U_e + U_B \tag{2.8}$$

This occurs when the total energy of the relativistic particles is close to equilibrium with the total magnetic field energy $U_B = \frac{3}{4}(1+k)U_e$, and therefore $U_{min} = \frac{7}{3}U_B$, and is often referred to as the equipartition value (Govoni and Feretti, 2004). This allows the calculation of the minimum energy density u_{min} which can be used to derive the associated equipartition field B_{eq}.

$$B_{eq} = \left(\frac{24\pi}{7}u_{min}\right)^{\frac{1}{2}} \tag{2.9}$$

However, the IC emission is independent of the magnetic field, meaning that comparisons between these fluxes would allow direct measurement of the magnetic field strength (Govoni and Feretti, 2004; Wik et al., 2014). Unfortunately, detection of the X-ray IC component is difficult as low X-ray energies are dominanted by the thermal ICM emission, while upper energies lack spatial resolution in current instruments. An interesting topic in its own right, IC emission and numeric estimations of cluster magnetic fields are outside the scope of this thesis.

2.2 Lifetime of synchrotron emission

The energy losses of the CRe as they interact with the electromagnetic fields and particles in the ICM eventually result in them no longer being energetic enough to produce observable radio emission. Considering losses due to synchrotron and IC processes, the lifetime of a CRe τ can be given as the following function of the magnetic field in the ICM B, the equivalent magnetic field of the CMB B_{CMB} (both fields are measured in μG), the cluster redshift z, and the observed frequency of the particle ν (van Weeren et al., 2019).

$$\tau = 32\frac{B^{\frac{1}{2}}}{B^2 + B_{CMB}^2}\left[(1+z)\nu\right]^{-\frac{1}{2}} \quad \text{[Gyr]} \tag{2.10}$$

As the majority of observed minihalos have redshifts in the range $[0.1, 0.3]$[2] and magnetic fields within the ICM are estimated at $\lesssim \mu$G, the maximum expected lifetime of a CRe is about 10^8 years (Sarazin, 1999; Feretti et al., 2012; Gitti et al., 2018).

A basic check for the spatial extent for these particles, in absentia of external forces $d \approx c\tau$, finds a distance of $\sim 10^{16}$ m or 1 pc from the source. This is orders of magnitude below the extents of observed diffuse radio emission, so we then consider that the CRe are moving within the bulk motion and magnetic fields of the ICM. By assuming that there is a strong enough turbulence to promote diffusion of CRe, we can use the Bohm approximation $D = \frac{k_B T}{eB}$, which models the diffusion of plasma across a magnetic field. Bagchi et al. (2002) finds that the expected length-scale of this diffusion (at $z = 0.3$) is that of a random walk across its lifetime

[2] A handful minihalos are notably outside this range, with the Phoenix cluster at $z = 0.596$ (van Weeren et al., 2014), ACTJ0022-0036 at $z = 0.8$ (Knowles et al., 2019), and a recent candidate SpARCS1049+56 at $z = 1.709$ (HLavacek-Larrondo et al., 2025)

with a diffusion coefficient κ and length-scale L_{diff},[3]

$$\kappa = 3.31 \times 10^{21} \left(\frac{E}{\text{GeV}}\right)\left(\frac{B}{10^{-1}\mu\text{G}}\right) \text{ m s}^{-1} \tag{2.11}$$

$$L_{diff} = \sqrt{\kappa\tau} = 6.7 \left(\frac{B}{10^{-1}\mu\text{G}}\right)^{-\frac{1}{2}} \text{ pc} \tag{2.12}$$

Once again, we find that the expected diffusion and halo produced by a point source is measured in the 10s of parsecs, far below the kiloparsec and megaparsec halos observed. It then seems a reasonable conclusion that these particles are instead reaccelerated or produced *in-situ* by a process that is extended over the kiloparsec scale and long-lived across the particle lifetime.

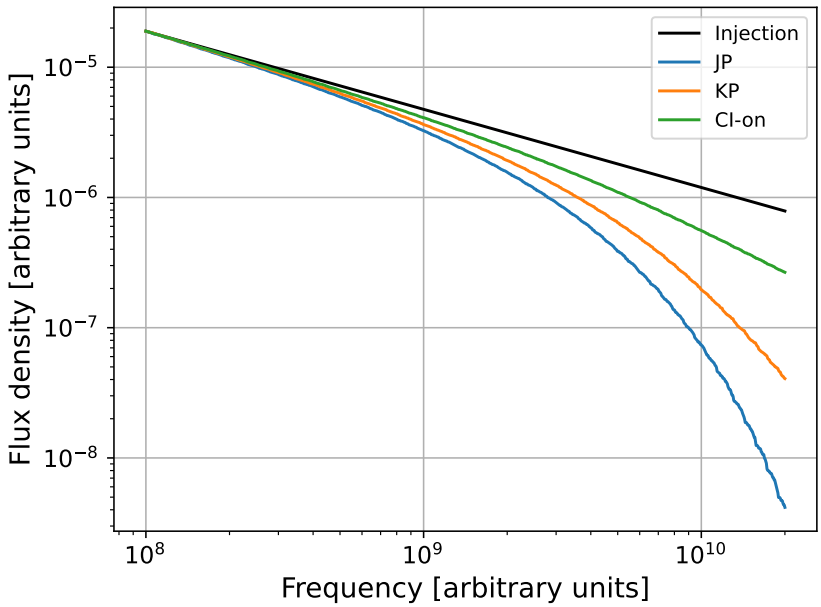

Figure 2.1: A representation of radio spectral shapes with $\alpha_{inj} = -0.6$. The injection spectrum depicts the shape prior to energy losses. The curves have been normalised to the injection spectrum at low wavelengths for better comparison.

The spectral shape of synchrotron emission is also affected by the lifetime of the particles, as higher energy particles have a shorter lifetimes ($\tau \propto \nu^{-\frac{1}{2}}$), causing the population of CRe change over time from the initial power-law. Several models have been developed to describe the resulting spectra, depending on the initial injection spectrum α_{inj}, break frequency, and any continuous injection after the initial state. The Jaffe-Perola (JP; Jaffe and Perola 1973) model assumes all CRe are created in a single burst, before aging with continuous scattering of electron pitch angles. This predicts that the pitch angle (between the electron velocity vector and local magnetic field vector) will be isotropic due to scattering. The Kardashev-Pacholczyk (KP;

[3]The length-scale diffusion is calculated as a random walk, scaling with the square root of time, rather than linearly.

Kardashev 1962, Pacholczyk 1970) model is similar to the JP model as it uses the same initial conditions, but evolves without pitch angle scattering. Continuous injection (CI) models describe sources that exist for longer than a single burst of acceleration, such as until current time (CI-on; Kardashev 1962), or for a fixed amount of time τ_{on} (CI-off; Komissarov and Gubanov 1994). An example of how these models differ in their shape is provided in Figure 2.1. The initial burst models show extreme steepening towards high frequency. In contrast, the CI model offsets the losses at high energy due to newly injected particles, but still has a lower level of steepening due to the presence of aged particles alongside newer ones.

The basic observables of synchrotron radiation can then be summarised in the following points.

- The synchrotron emissivity is dependent on the number density of the CR electrons and the strength of the magnetic field (Equation 2.7). Assuming a constant magnetic field strength, the emissivity follows a power-law $\epsilon_s(\nu) \propto \nu^{-\alpha}$ which is related to the CRe energy spectral index δ by $\alpha = (\delta - 1)/2$. Typical minihalo spectral indices are steep with $\alpha > 0.7$.[4]

- As the emitting particles lose energy the particle energy distribution changes, and therefore, the emitted radio spectrum changes. This appears as a steepening or curved spectrum at high frequencies, rather than a single power-law. The break frequency at which this occurs is related to the time since acceleration. This means that radio events that have stopped will show an aged spectrum with a characteristic break, i.e. radio relics. Ongoing events that are continually injecting new particles will not feature the same level of steepening.

2.3 Production mechanisms

With this understanding of the spectrum of CRe, we now consider the methods that can produce them. This can be broken down into primary models involving first and second order Fermi acceleration, and secondary models where the electrons are a byproduct or resulting from decay.

2.3.1 Turbulent reacceleration

This process takes a pre-existing lower energy CRe population of $\gamma \sim 100$ and boosts it to the factors observed $\gamma \gtrsim 1000$. This occurs via interactions with plasma waves and shocks within the ICM which can be caused by several factors, such as violent cluster mergers and gas sloshing within the core (Brunetti et al., 2001; Fujita et al., 2004; Mazzotta and Giacintucci, 2008).

Shocks within the ICM can cause first order Fermi acceleration (Fermi-I), also known as diffuse shock acceleration (DSA). As particles cross back and forward over the shock front, they can be reflected by magnetic instabilities upstream and downstream from the shock. Each

[4] A typical radio galaxy emits synchrotron radiation with a spectral index ~ 0.7.

crossing of the front increases the particle energy, raising trapped particles to relativistic levels before they escape. The energy gained via Fermi-I acceleration from a shock with velocity v_{wave} scales, compared to the speed of light c, as

$$\Delta E \propto \frac{v_{wave}}{c} E \tag{2.13}$$

Shocks are generally created due to merger activity in disturbed clusters, not relaxed clusters where minihalos are found. As such, it is not considered a major mechanism at play in the context of minihalos.

Gas sloshing within the cool core of relaxed clusters has been shown as a natural consequence of the larger motions of the ICM (Fujita et al., 2004). This turbulent environment naturally allows for low relativistic 'seed' electrons to undergo second order Fermi (Fermi-II) acceleration. This type of acceleration is stochastic as particles are scattered by the magnetic instabilities from the randomly moving plasma waves. Particles meeting these waves head-on are reflected with an increased energy, while particles in a head-tail collision are reflected with a lower energy. The chance for a head-on collision is higher than that of a head-tail collision allowing for a net gain. The mean energy gained is dependent on the speed of the waves and energy of the seed particle,

$$\Delta E \propto \left(\frac{v_{wave}}{c}\right)^2 E \tag{2.14}$$

The random nature of Fermi-II acceleration makes for an inefficient process compared to the aligned shock fronts of Fermi-I. Since $v_{wave} \ll c$, Fermi-II scales far more slowly than Fermi-I $(v_{wave}/c)^2 \ll v_{wave}/c$.

Brunetti and Lazarian (2007) show that reacceleration of relativistic particles moving through magnetohydrodynamic (MHD) turbulence are primarily accelerated by two Fermi-II type mechanisms.

The first mechanism is the damping of fast magnetosonic waves due to resonant transit time damping (TTD). The compressible and low-frequency plasma waves can resonate with the naturally gyrating particles at the zeroth order, $\omega = v_{\|} k_{\|}$, where ω is the frequency of the wave, $k_{\|}$ is the wavenumber, and $v_{\|}$ is the speed of the particle along the axis of the magnetic field. The random accelerations can then be captured as a random walk in momentum space, giving rise to a momentum-diffusion coefficient $D_{pp,TTD}$. ZuHone et al. (2013) simulate this process numerically in the case of minihalos and simplify this diffusion coefficient to be proportional to the average wavenumber $\langle k \rangle$, the turbulent velocity of the particle v_t,[5] and fraction of turbulent energy in compressible motion R_c.

$$D_{pp,TTD} \propto \langle k \rangle v_t^2 R_c p^2 \tag{2.15}$$

The second mechanism is through the non-resonant compression and expansion of the ICM. Adiabatic compression increases the particle momentum, while expansion reduces it. As with

[5]This is essentially the instantaneous velocity of the particle due to turbulence.

collisions, the particle is more likely to experience compression than expansion, leading to a net gain in energy. Here ZuHone et al. (2013) take the fast diffusion limit, where particles escape the eddies of compression before they re-expand. This non-resonant momentum-diffusion coefficient scales as

$$D_{pp,C} \propto \frac{V_l^2}{D} p^2 \tag{2.16}$$

where V_l is the compressive velocity at scale l, and $D \propto c l_{mfp}$ is the spatial diffusion rate for a mean free path of l_{mfp}.

The reacceleration efficiency is then given via $\chi = \frac{4D_{pp}}{p^2}$, where D_{pp} is the sum of the prior diffusion coefficients. The total rate of momentum increase for each particle can then be clearly written as the sum of two Fermi-II processes when compared to Equation 2.14.[6]

$$\frac{dp}{dt} = \chi p \tag{2.17}$$

$$\frac{dp}{dt} \propto \langle k \rangle R_c v_t^2 p + \frac{1}{D} V_l^2 p \tag{2.18}$$

One remaining loose thread in this explanation is the source of the seed electrons themselves. A possible explanation is that the seed population is provided from radio AGN and spread throughout the cluster core by the gas sloshing (Brunetti et al., 2001). The CRe from the central BCG lose energy and disappear out of our detectable range by the physics described earlier, however, they don't stop existing after this timescale. These 'relic' CRe are lower energy but still relativistic with $\gamma \sim 100$, radiating at radio frequencies that we are unable to detect. In this state they still continue to move and disperse through the core, creating the seeds for the large scale minihalos that we observe. The central BCG may not be the only source, other AGN within the cluster, galactic winds, supernovae, may all contribute relic CRe to be later reaccelerated. ZuHone et al. (2013) find that the sloshing should be sufficient to redistribute these particles to create a minihalo. Hadronic interactions as described in a later section could also contribute seed electrons.

2.3.2 Observables of turbulent reacceleration

With this understanding we can now explore the spectral and spatial features that we expect to observe from emission resulting from the reacceleration scenario. Turbulence is an inherently transient and intermittent phenomenon. This spatial variance means that when we sample a given region of space, we may instead be observing several 3D components superimposed on our 2D image. Areas with higher turbulent velocity inspire greater magnetic fields and greater levels of emission (ZuHone et al., 2013). The lack of a steady state and variable components means that we might expect variation in the flux brightness and spectral index across different locations of

[6]The factors of c have been absorbed into the constants, with the main focus being the squared scaling with velocity and linear with momentum

the minihalo (Riseley et al., 2022). However, due to the integration over the line of sight, these artefacts could be averaged and less visible in the 2D observation.

The particles must also trace the distribution of turbulence within the ICM, leading to a spatial bounding of the minihalo within the gas sloshing of the core. The boundary of the cluster core is often associated with cold fronts visible in X-ray data, allowing comparison between X-ray and radio images. Cold fronts denote the boundary between cooler denser gas clouds moving through a hotter and more diffuse surrounding gas. These are notably different to shock fronts, where the boundary separates a hot and dense downstream region that the shock has passed through, from the cooler and less dense upstream region the shock is moving towards (Markevitch and Vikhlinin, 2007). ZuHone et al. (2013) find that simulated turbulent re-acceleration is indeed bounded by the cold fronts (Figure 2.2), with a sharp steepening at the core radius. They also find that depending on the projection through the ICM, it may be difficult to locate the cold fronts in X-ray images due to the afore mentioned line of sight integration.

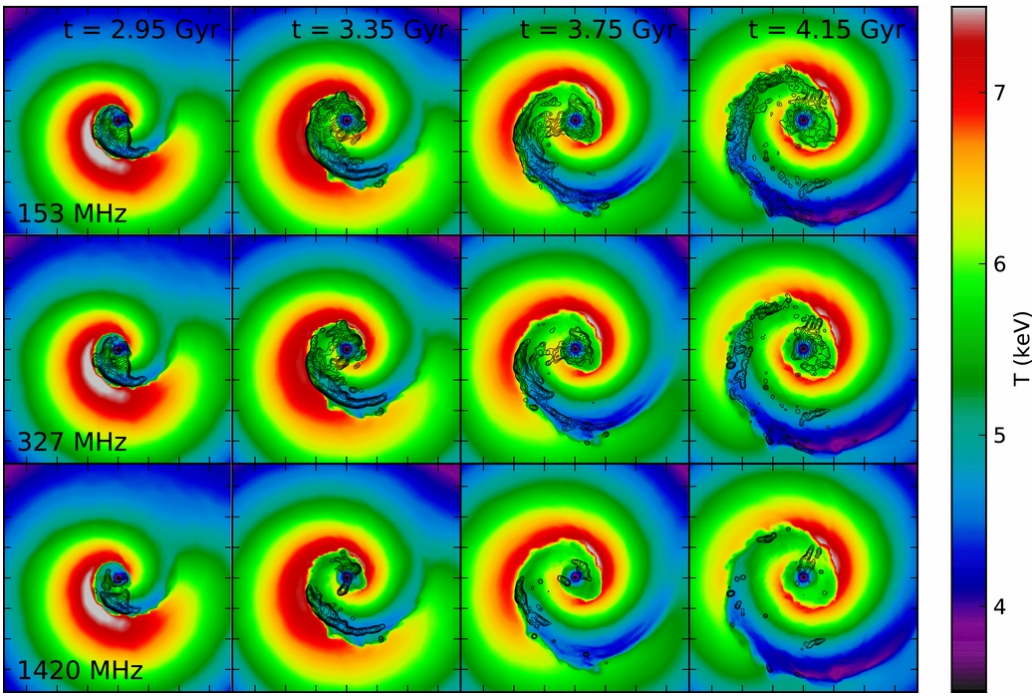

Figure 2.2: 2D projected gas temperature maps with overlaid radio contours at several simulation timesteps for frequencies 153, 327, and 1420 MHz. Radio emission contours begin at $(1.0, 0.5, 0.125) \times 10^{-3}$ mJy arcsec^{-2} and increase by a factor of 2. Image dimensions are 750×750 kpc, with tick marks every 100 kpc. (Figure 10 of ZuHone et al. (2013))

A spectral feature of the reacceleration model arises from the balancing of reacceleration and losses, creating a cutoff energy at which the population of CRe declines more sharply. This is indeed the logic behind the CI model visualised in Figure 2.1, with a steepening in the spectral index beyond the break frequency. ZuHone et al. (2013) note that as there is a range of magnetic field strengths within the core, this break frequency will change between CRe, causing a smearing of the break and a more gradual steepening towards high frequency.

2.3.3 Hadronic Production

For the secondary production we instead consider the protons within the ICM, rather than the electrons. We again separate them into a relativistic 'cosmic ray' proton (CRp) population and a thermal population and examine the resultant interactions. CRp have significantly longer radiative lifetimes than CRe, with $\tau_p \sim 10^{10}$ yrs, allowing them to travel up to 1 Mpc (Dennison, 1980). The spread out CRp (p_{CR}) then collide inelastically with thermal protons (p_{th}) to produce charged and neutral pions π

$$p_{CR} + p_{th} \rightarrow \pi^0 + \pi^+ + \pi^- \tag{2.19}$$

the charged pions then decay into (anti)muons $\mu^\pm$ and muon (anti)neutrinos $\nu_\mu(\tilde{\nu}_\mu)$, while the neutral pions decay into gamma rays γ

$$\pi^\pm \rightarrow \mu^\pm + \nu_\mu(\tilde{\nu}_\mu) \tag{2.20}$$

$$\pi^0 \rightarrow 2\gamma \tag{2.21}$$

the muons then decay into highly relativistic (positrons)electrons $e^\pm$ and electron (anti)neutrinos $\nu_e(\tilde{\nu}_e)$.

$$\mu^\pm \rightarrow e^\pm + \nu_e(\tilde{\nu}_e) + \tilde{\nu}_\mu(\nu_\mu) \tag{2.22}$$

From this chain reaction we end up with two observables, CRes and gamma rays. Importantly, it is these secondary CRe products that are predicted to produce the minihalo emission. The emission in this model then traces the population of CRp producing the CRe, rather than the turbulence of the ICM. Due to our assumption of the long radiative lifetimes, we expect a continuous injection of CRe from the hadronic process.

ZuHone et al. (2015) investigate the hadronic method via simulation and assume a power-law momentum distribution for the CRp, $n_p(p_p) = C_p p_p^{-\alpha_p}$ (where C_p is a normalisation constant), and estimate the CRp momentum distribution spectral index to be in the range $\alpha_p \sim 2.0$–2.5. They then derive a distribution for the produced CRe momentum p based upon the CRp momentum distribution to be

$$Q(p)dp = \frac{4}{3} 16^{1-\alpha_p} c n_{th} C_p \sigma_{pp}(16p) p^{-\alpha_p} dp \tag{2.23}$$

with n_{th} the density of thermal protons, $\sigma_{pp}(16p)$ the proton-proton inelastic collision cross section at 16 times the CRe momentum.[7] Ignoring the numerical terms we can also write

$$Q(p)dp \propto n_{th} \sigma_{pp} p^{-\alpha_p} dp \tag{2.24}$$

to clearly see that the efficiency of the hadronic method scales with the density of thermal protons, collision cross section, and as a power-law in momentum. By approximating σ_{pp} as constant[8]

[7] This is an approximation of the CRp momentum in the proton-proton collision, $p_p = 16p$.

[8] There is a weak dependence on the CRp and thermal proton energies, which ZuHone et al. (2015) account for, resulting in a minor flattening effect on the synchrotron spectral index.

we are able to find a steady state for the number density of CRe by equating the synchrotron and IC losses $\frac{dp}{dt} \propto -p^2$. This occurs when $\frac{d}{dp}(-p^2 n_e) + Q(p) = 0$, which following through by integrating for all particles $p \to \infty$,

$$
\begin{aligned}
-p^2 n_e &= -\int_p^\infty Q(p) \\
p^2 n_e &\propto \int_p^\infty p^{-\alpha_p} \\
n_e &\propto \frac{p^{-\alpha_p+1}}{p^2} \\
n_e &\propto p^{-(\alpha_p+1)}
\end{aligned}
$$

This gives us a CRe density power-law with index $\delta = \alpha_p + 1$ and therefore a synchrotron spectral index of $\alpha = \frac{\alpha_p}{2}$. Using the expected values for α_p we then should expect $\alpha \sim 1$–1.25 which is similar to that of observed minihalos. However, this assumes steady state injection which is a somewhat idealised case for the intermittent and turbulent core of the cluster.

2.3.4 Observables of hadronic production

This hadronic process predicts a flat spectral index extending into high frequencies, lacking the steepening observed in the reacceleration model. This also implies the spectral index will be constant throughout the spatial extent of the minihalo, rather than varying due to localised high amounts of turbulence. Keshet and Loeb (2010) show that changes in the magnetic field strength relative to B_{CMB} can induce changes in the emissivity. For regions of the magnetic field where $B < B_{CMB}$ they predict a radio suppression $\propto (B/B_{CMB})^2$. Assuming that the magnetic field inside a cluster has a radial steepening, there then exists a radius r_b where the magnetic field drops below B_{CMB} and a radial break may be observed in the radio profile. As such, the extent of the minihalo would be more constrained via the strength of the magnetic field, than the cold fronts of the cool core.

The simulation of hadronic production by ZuHone et al. (2015) find that the majority of particles have spectral indices close to that of steady state. As B decreases towards $B \ll B_{CMB}$ the spread of spectral indices converges tightly. The opposite happens at $B \gg B_{CMB}$ with broadening and a small skew towards steeper indices. They attribute the broadening to sampling regions with rapidly decreasing magnetic fields (flatter index) and rapidly increasing magnetic fields (steeper index). They also note that the spatial bounding of the hadronic emission is less constraining than that of the reacceleration model. A drop in the emission across the cold front occurs due to a drop in the magnetic field strength and falls $\propto B^2$, but they simultaneously observed emission extending outside of the cool core.

Importantly, ZuHone et al. (2015) do not find evidence of a spectral flattening at high frequencies, showing that a simple power-law relationship holds to beyond 10 GHz. They

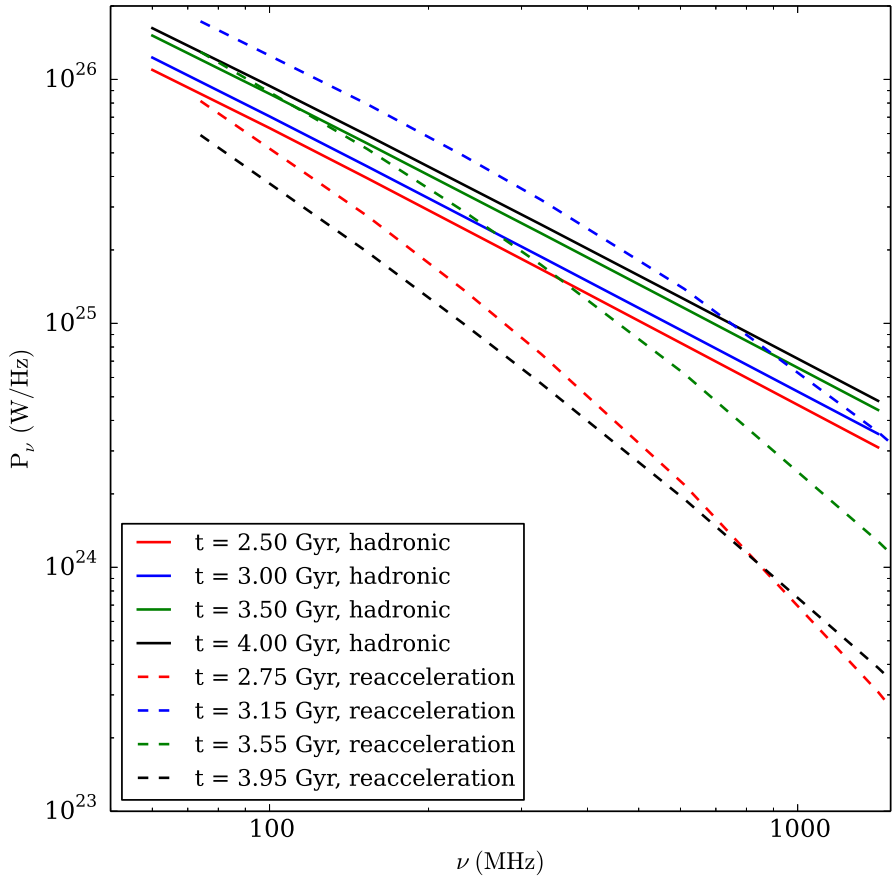

Figure 2.3: The total simulated radio spectrum within 300 kpc radius of the cluster centre, comparing the hadronic and reacceleration model at several timesteps. (Fig 8 of ZuHone et al. 2015)

compare the spectral distribution of the hadronic and reacceleration models and clearly show the difference in spectral shape (Figure 2.3).

A final clear distinction from the reacceleration model is the emission of gamma rays with the hadronic model. This implies that there will be a diffuse gamma emission aligned spatially with the minihalo. This would be a 'smoking gun' proof for the hadronic model, however as mentioned earlier, current instruments don't have the spatial resolution to accurately detect diffuse gamma emission at minihalo scales.

2.4 Previous minihalo studies

Minihalos are the most poorly understood class of diffuse radio emission, with few in-depth spectral studies to look at. For many minihalos we only have a couple of observations between 144 MHz and 1.4 GHz taken by the *Giant Metrewave Radio Telescope* (GMRT) and VLA (Giacintucci et al., 2014b, 2019). This lack of data means that for many minihalos the spectral index is calculated from a few points, if it can be at all. Furthermore, the lack of high frequency

data > 1.4 GHz makes detecting steepening from the reacceleration theory far more challenging, with both models resembling a power-law within this range (Figure 2.3).

The most in-depth spectral study at a range of frequencies has been performed on RXJ1720+2638 (Mazzotta and Giacintucci, 2008; Giacintucci et al., 2014a; Perrott et al., 2021). This revealed a tail to the minihalo, with a steeper spectrum than the central region. Early analysis showed a potential for steepening at 5 GHz, however, later observations at 13-18 GHz were consistent with a power-law index $\alpha = 1.0$ (Perrott et al., 2021). These spectral features lend impetus towards the hadronic scenario for the core, while also suggesting that reacceleration might play a role within the tail. The tail of the minihalo also appears to be constrained by the cold fronts of the core. More detailed analysis for this cluster is given in Section **??** , which includes images produced as part of this thesis.

Fix reference once we get there

The recent LOFAR-meets-MeerKAT series (Riseley et al., 2022, 2023, 2024) aims to perform a more uniform sample of minihalo observations to investigate the method of CRe production. These observations are performed at 145 MHz to 1.7 GHz with the afore mentioned LOFAR and MeerKAT telescopes and used alongside archival *Chandra* X-ray data. These detailed analyses of MS1455+2232, A1413, and A2142 have revealed that the minihalo morphology generally contains multiple components, a single steep-spectrum power-law, and spatially homogeneous spectral index. They find that both scenarios can explain several observables, but not all, and suggest that both mechanisms might be at play. MS1455+2232 and A1413 are discussed in more detail, alongside images produced during this thesis, in sections **??** and **??** respectively.

Fix reference

Another point of interest is in a possible connection to giant radio halos. Although minihalos are found exclusively within cool-core clusters and giant radio halos are found almost entirely within disturbed clusters, they may have a physical relation to each other. The general consensus is that giant halos are formed by reacceleration due to merger-induced turbulence, due to a lack of gamma emission at the larger cluster scale (Brunetti et al., 2012; Brunetti et al., 2017). These two phenomena may have a transitional state where a minihalo switches into a giant halo due to a merger, and the inverse as the cluster relaxes (Brunetti and Jones, 2014). An alternative scenario could be that mini-halos are produced via the hadronic process, which shifts towards turbulent reacceleration at larger scales (Zandanel et al., 2014).

The cluster A2142 shows some evidence towards these theories, with a large scale 2 Mpc halo comprised of two parts (Venturi et al., 2017). This cluster is undergoing a minor merger and has only small disturbances in the ICM. The brighter inner core component has a spectral index $\alpha_i = 1.33 \pm 0.08$, which appears to be produced similarly to that of a minihalo. The outer component is fainter and somewhat steeper with $\alpha_o \approx 1.5$, suggested to be caused by large scale turbulence within the ICM. Clusters may also show evidence of a minihalo and giant halo existing simultaneously, with RXJ1720+2638 and PSZ139.61+24.20 both exhibiting large scale steep spectrum emission outside the core region (Savini et al., 2018, 2019). Physical relations

to giant radio halos are useful in our understanding of how mergers change the behaviour of a cluster, but are outside the scope of this thesis.

2.5 Interferometry

To better understand the reasoning behind the calibration and imaging processes, it is useful to first cover the basics of interferometry. Interferometer arrays consist of individual antennas that measure the amplitude and phase of incoming electromagnetic waves. These signals are then combined with each other to produce an interference pattern. The simplest example for this is an array of two antennas. Since each antenna has a small difference in distance from the source, there is a geometric time delay τ_g between each receiving a signal. By combining the two individual signals, the interference pattern is created due to the phase shift caused by the time delay. Given a cosinusoidal wave travelling at the speed of light c, from a source in the far field at $\boldsymbol{s}$, and the baseline distance between the two antennas $\boldsymbol{b}$, the geometric delay is $\tau_g = \boldsymbol{b} \cdot \boldsymbol{s}/c$. The measured interference pattern is then the product of the individual fields $V_1 = E\cos(\omega t)$ and $V_2 = E\cos(\omega(t-\tau_g))$, which can be written in the form $\frac{E^2}{2}\left[\cos(\omega\tau_g) + \cos(2\omega t - \omega\tau_g)\right]$ (Perley, 2018). The second term varies much faster than the first and can be averaged out, and the prefactor can be shortened to $P = \frac{E^2}{2}$. This gives the final cosine response as

$$R_C = P\cos(\omega\tau_g) \tag{2.25}$$

or written in terms of the signal wavelength $\lambda = 2\pi c/\omega$ and the baseline,

$$R_C = P\cos\left(2\pi\frac{\boldsymbol{b}\cdot\boldsymbol{s}}{\lambda}\right) \tag{2.26}$$

The fringes of the interference pattern then sample the sky, with longer baselines corresponding to small spatial features, and small baselines mapping large spatial features. For the sky brightness map $I_\nu(\boldsymbol{s})$ at the frequency ν, the interferometer response is the spatial integral[9] of the sky modified by the cosinusoidal fringe pattern.

$$R_C = \iint I_\nu(\boldsymbol{s})\cos\left(2\pi\nu\frac{\boldsymbol{b}\cdot\boldsymbol{s}}{c}\right)d\Omega \tag{2.27}$$

This response function is even due to the cosine term, resulting in any odd functions described by the intensity map being averaged out. By modifying the phase of one of the antennas by 90 degrees, $V_2 = E\sin(\omega(t-\tau_g))$, and applying the same processes as before, an odd response function is created $R_S = P\sin(\omega\tau_g)$. The two responses R_C and R_S can then be used to define the complex visibility function in terms of $\boldsymbol{u} = \boldsymbol{b}/\lambda$ (Perley, 2018).

$$V(\boldsymbol{b}) = R_C - iR_S = \iint I_\nu(\boldsymbol{s})e^{-2\pi i\boldsymbol{u}\cdot\boldsymbol{s}}\,d\Omega \tag{2.28}$$

[9]The spatial integral is in units of solid angle Ω, as the usual axes of the sky are given in arcseconds, or arcminutes.

If the baselines lie on a flat plane, the complex visibility is then a 2D Fourier transform of the sky brightness at the observing frequency, allowing an inverse transform to be performed to reconstruct the sky given complete sampling. The baseline vectors $\boldsymbol{u} = (u, v, w)$ are defined in units of wavelength, with the flat uv-plane defined as a special case having $w = 0$. In 3D coordinates, the unit vector $\hat{\boldsymbol{u}}$ is aligned East-West, $\hat{\boldsymbol{v}}$ is North-South, and $\hat{\boldsymbol{w}}$ points towards the source (or phase centre). If the antenna array is not coplanar to the source, baselines may vary in w, which invalidates the 2D Fourier transform. To solve this, many imaging packages project the visibilities onto the uv-plane (called w-projection) (Perley, 2018).

An individual antenna pair only samples a single point in $\boldsymbol{u}$, so to maximise coverage of the uv-plane, interferometer arrays use N antennas in tandem to achieve $N(N-1)$ unique pairs simultaneously. Similarly, the rotation of the Earth allows these baselines to move with respect to the source, sweeping through the uv-plane in an ellipse to increase coverage. As source declination approaches 0°, the ellipse flattens due to the source alignment with the Earth's rotation, reducing effective coverage. The final coverage of the uv-plane then depends on the arrangement and number of antennas, the location in the sky that is being observed, and the length of time that the source is being observed.

The incomplete coverage of the uv-plane results in a synthesised beam with radial spikes and periodics in image space, called the point spread function (PSF). The inverse transform of the sampled complex visibilities is then called the 'dirty' image, having the true sky convolved with the PSF. Several methods have been developed to deconvolve the PSF, with the most common method being the CLEAN algorithm (Högbom, 1974; Clark, 1980). This follows an iterative major-minor cycle to build a model of the sky brightness. Minor cycles operate in the image space, identifying source components in the residual image and adding to the sky model. The major cycle transforms the model image into visibilities and subtracts them from the observed visibilities. These residuals are then Fourier transformed to update the residual image. With tuned parameterisation, the CLEAN algorithm can successfully deconvolve the PSF and return an accurate image of the sky brightness.

In practice, noise and calibration errors add further complexity to the observed complex visibilities. These can be delay errors, antenna-dependent gain and phase variations, atmospheric effects, and instrumental noise. This can be summarised via the Measurement Equation, which defines the observed visibilities $\boldsymbol{V}_{ij}$ between antennas i and j in terms of the ideal visibilities $\boldsymbol{V}_{ij}^{IDEAL}$ and an accumulation of corrupting effects $\boldsymbol{J}_{ij}$ for that baseline (Hamaker et al., 1996; Sault et al., 1996).

$$\boldsymbol{V}_{ij} = \boldsymbol{J}_{ij}\boldsymbol{V}_{ij}^{IDEAL} \tag{2.29}$$

The term $\boldsymbol{J}_{ij}$ can be factored into a sequence of individual effects applied to the incoming wave, with different physical origins and importances. Antennas measure the polarisation of the incoming wave, forming four correlations in a circular basis: LL, RR, LR, RL. As a result,

the matrices of the Measurement Equation are 2×2 to describe the polarisation response. In the context of minihalos, the cross-polarisations are not of scientific interest and are used only for calibration purposes. The specific calibration and imaging procedures applied to the data analysed in this thesis are presented in the following chapter.

Chapter 3

Calibration and Imaging

The observations to be calibrated are VLA X-band (8–12 GHz) data, taken in C and D configurations of 12 galaxy clusters hosting minihalos. VLA antennas may be arranged in four standard configurations, A through D. A has the longest baselines and best angular resolution, at the expense of lower brightness sensitivity. Baseline length decreases towards D-configuration, which has the poorest angular resolution, but greatest brightness sensitivity. D-configuration was chosen to maximise brightness sensitivity to detect the faint and extended minihalos. C-configuration observations were added to allow accurate imaging of compact AGN sources embedded within the minihalos.

The C-configuration observations were performed between January and February 2024, under project code 24A-411 (PI: Yvette Perrott). The D-configuration observations were taken the following year during February and March 2025 under project code 25A-157 (PI: Yvette Perrott). The frequency range for X-band spans 7.977–12.025 GHz, with a total bandwidth of 4.048 GHz. This is split into 32 spectral windows with individual bandwidths of 128 MHz. Each spectral window has a spectral resolution of 2 MHz, resulting in a total of 64 channels per window. For reference, an overview of the targeted clusters is given in Table 3.1, while details of the radio observations of each cluster are given in Table 3.2.

Add max-min baselines

To calibrate and image these observations, *Common Astronomy Software Applications* (CASA) has been developed by an international collaboration of scientists and software developers.[1] This provides Python-based tools for the processing, analysis, and manipulation of observations from the VLA and the Atacama Large Millimeter/submillimeter Array (ALMA). Part of this software suite is the VLA pipeline, which is able to perform calibration and automated flagging of raw observations. The pipeline allows a streamlined process with reduced manual intervention, while producing a final weblog detailing the applied steps for reproducibility. Flagging of data refers to the exclusion of problematic points, often due to the presence of

[1] https://casa.nrao.edu/index.shtml

Cluster	R.A.$_{J2000}$ (h, m, s)	Decl$_{J2000}$ (deg, arcmin, arcsec)	z	Scale (kpc arcsec^{-1})
2A0335+096	03 38 40.5	+09 58 12.4	0.036	0.716
A2626	23 36 30.5	+21 08 47.6	0.055	1.069
A1795	13 48 52.5	+26 35 34.0	0.062	1.195
A478	04 13 25.3	+10 27 55.0	0.086	1.612
A1413	11 55 18.0	+23 24 19.0	0.141	2.482
A2204	16 32 47.0	+05 34 35.0	0.152	2.643
RXJ1720+2638	17 20 10.0	+26 37 32.0	0.160	2.758
RXJ2129+0005	21 29 39.9	+00 05 21.3	0.234	3.722
MS1455+2232	14 57 15.1	+22 20 34.0	0.257	3.990
Z3146	10 23 39.6	+04 11 11.0	0.288	4.329
RXCJ1115+0129[a]	11 15 52.0	+01 29 54.0	0.350	4.940
ACTJ0022-0036[b]	00 22 13.0	−00 36 33.0	0.804	7.521

Notes. Right ascension and Declination values are taken from the observation phase centres. Redshifts are from the NASA/IPAC Extragalactic Database (NED). Values for kpc arcsec^{-1} are calculated using the Cosmology Calculator I (Wright, 2006).
[a] Also known as ZwCl 1454.8+2233.
[b] Also known as ZwCl 1113.8+0144.

Table 3.1: List of observed clusters, sorted by redshift.

radio frequency interference (RFI). For these observations, pipeline flagging did not sufficiently mitigate the RFI for accurate calibration, necessitating the development of a modified version. A description of the pipeline steps and calibration is provided in Section **??**. The following section details the RFI present in the data and the measures taken to mitigate it.

3.1 Radio Frequency Interference

An unavoidable consequence of ground-based antennas is that they can pick up spurious radio sources, broadly termed RFI. This can be attributed to internal mechanisms of the telescope, as well as external sources such as ground-based and satellite communications. With the increasing number of satellites in orbit, RFI is a growing problem, with only a few narrow frequency ranges protected for radio astronomy. RFI is particularly strong near the declination of 0°, as this aligns with the orbit of geostationary satellites, known as the Clarke Belt. RFI can be categorised as intermittent or continuous in time, and as broadband or narrowband in frequency. Detecting and automatically removing these data can be difficult due to the wide range of forms RFI can take. NRAO monitors RFI across the VLA frequency bands and provides descriptions of known sources to aid in cleaning the data. For observations at X-band,[2] satellite and weather radars are an intermittent source between 9.3–9.9 GHz, continuous ground-based microwave links contaminate 10.7–11.5 GHz, and strong RFI exists between 11.7–12 GHz due to emission

Image of interfernce 2D?

[2] https://science.nrao.edu/facilities/vla/observing/RFI/X-Band

Table 3.2: Summary of the VLA X-band observations and calibration sources used.

Cluster	VLA Config.	Date	Time (min)	Primary Calib.	Secondary Calib.
2A0335+096	C	2024-01-27	45.3		
	D	2025-02-28	68.5	3C48	J0321+1221
	D	2025-03-15	68.5		
A478	C	2024-01-27	44.5		
	D	2025-02-27	124.5	3C48	J0424+0204
	D	2025-03-11	124.5		
A2204	C	2024-02-03	64		
	D	2025-02-28	36.6	3C286	J1616+0459
	D	2025-03-27	36.6		
MS1455+2232	C	2024-02-03	61.1		
	D	2025-03-19	36.2	3C286	J1436+2321
	D	2025-03-27	35.6		
RXJ1720+2638	C	2024-02-03	62.2		
	D	2025-02-28	35.6	3C286	J1719+1745
	D	2025-03-27	35.6		
A2204	C	2024-02-03	64		
	D	2025-02-28	36.6	3C286	J0424+0204
	D	2025-03-27	36.6		
A1413	C	2024-02-04	36.9		
	D	2025-02-28	129.2	3C48	J1150+2417
	D	2025-02-28	129.2		
A1795	C	2024-02-04	71.5		
	D	2025-03-19	35.6	3C286	3C286
	D	2025-03-27	35.9		
RXCJ1115+0129	C	2024-02-04	73.1		
	D	2025-02-27	36.5	3C286	J1058+0133
	D	2025-03-16	36.5		
Z3146	C	2024-02-04	73.1		
	D	2025-02-27	36.5	3C48	J1041+0610
	D	2025-03-16	36.5		
A2626	C	2024-02-10	41.5		
	C	2024-02-11	41.5	3C48	J2253+1608
	D	2025-02-28	41.5		
	D	2025-02-28	33.2		
ACTJ0022-0036	C	2024-02-10	109.5	3C48	J0022+0014
	C	2024-02-11	109.5		
	D	2025-02-28	27.4		
	D	2025-02-28	36.5		
RXJ2129+0005	C	2024-02-10	138.1	3C48	J2136+0041
	C	2024-02-11	138.1		
	D	2025-02-28	64.5		
	D	2025-02-28	64.5		

Notes. The primary calibrator was used for setting the flux scale and bandpass calibration. The secondary calibrator was used for phase calibration.

associated with the Clarke Belt.

For the faint sources of minihalos ($\sim \mu$Jy), RFI signals can be orders of magnitude brighter (mJy). This smothers the signal and can bias calibration results. Sources of RFI may have different delays in reaching antenna compared to the science target, which can bias delay and phase calibrations and increase noise. This degrades the angular and brightness sensitivities. When imaging, RFI may appear as bright wave-like artefacts, arising from the Fourier transform of high-amplitude visibilities with random phase.

Ref for algorithms

The standard pipeline flagging uses a combination of the tfcrop and rflag algorithms to detect and remove RFI. tfcrop uses the visibility amplitudes to form an average time spectrum, and a polynomial model for the frequency bandshape. Using this model, it then flags outlier points beyond a given relative threshold. This is adept at flagging short-duration spikes of RFI in both narrowband and broadband ranges, as well as time-persistent RFI in the narrowband. rflag uses a sliding window to calculate the local visibility RMS, and flags points beyond a relative threshold. This operates on individual baselines over single frequencies (channels) with the sliding window over the time axis, followed by single time steps with the sliding window across channels. rflag is best at identifying weak and sparse RFI that may be affecting a subsection of antennas. The pipeline also extends the flags for each baseline, such that any channel or time step that is > 50% flagged will be entirely flagged. Flags are also propagated between polarisations, with all four correlations flagged if any individual polarisation is flagged.

To improve upon the results from the algorithms implemented in CASA, I tested `AOFLAGGER`, which was originally designed for LOFAR but has since expanded for use with several other telescopes (Offringa et al., 2010, 2012, 2023).[3] This is implemented via a flexible Lua script, allowing customisation in algorithms to each telescope and observation setup. The default VLA script performs RMS-based time step and channel flagging, high-pass filtering, and morphological filtering. The morphological analysis is performed via the `SumThreshold` algorithm, which uses several decreasing thresholds to evaluate a sequence of samples. Points exceeding the highest threshold are marked for flagging, after which next lower threshold is evaluated, and the process is repeated. This works well for broadband and peaked RFI. This script has two primary parameters, `base_threshold` and `transient_threshold_factor` which define the thresholds to use for flagging. Lower thresholds correspond to increased sensitivity, and therefore more aggressive flagging. For frequent but intermittent RFI bursts, `transient_threshold_factor` can be used to tune detection separately to the main threshold. The script was altered to preserve any pre-existing flags applied by the pipeline.

Based on the NRAO description of RFI in the X-band, the script is configured to do three passes with separate parameters for each. The first two passes flag only 9.3–9.5 GHz (weather radar) and 10.7–12 GHz (ground communications), before a final full frequency range pass. This

[3] https://aoflagger.readthedocs.io/en/latest/index.html

allows for tuning of aggression based upon the RFI present in each observation. The higher angular resolution C-configuration observations had a much lower amount of RFI compared to the D-configuration, where many datasets were sufficiently contaminated by the ground communications that these spectral windows were completely removed. A summary of the final datasets is given somewhere. Scripts were tuned for each source and observation independently using a manual and iterative testing process. An example Lua script is given Appendix A. **Need to do this**
This process was incorporated into a copy of the VLA Pipeline version 6.6.1 as a custom step `hifv_aoflagger` with three arguments: the source target to be flagged `flag_target`, the lua script `aoflagger_file`, and the data column to use (raw or corrected). If the corrected data column is selected, this first subtracts the model from the corrected data to estimate the residual spectrum. Given an accurate source model, these residuals are expected to be closer to a Gaussian noise distribution, improving RMS-based RFI detection. This provides flexibility per observed source with customisable parameters per source and region of RFI. The implementation of this custom pipeline step is available in a separate repository.[4]

3.2 Calibration pipeline

All observations were calibrated using a single Python file, `casa_pipescript.py`. An example pipeline script is given in Listing 3.1. The following descriptions are written with reference to the Pipeline Tasks Reference Manual and CASA docuumentation.[5,6].

The first stage `hifv_importdata` is to loads the data, in the format of a Measurement Set (MS), with the associated metadata. This context is used throughout the pipeline to determine which scans, sources, and spectral windows are evaluated.

Listing 3.1: Example CASA pipeline script.

```
hifv_importdata(vis=["DATASET"])
hifv_flagdata(
    hm_tbuff='1.5int', fracspw=0.01, intents='*POINTING*,
    *FOCUS*,*ATMOSPHERE*,*SIDEBAND_RATIO*, *UNKNOWN*,
    *SYSTEM_CONFIGURATION*, *UNSPECIFIED#UNSPECIFIED*')
hifv_vlasetjy()
hifv_priorcals(show_tec_maps=False)
hifv_aoflagger(
    flag_target='PRIMARY', aoflagger_file='PRIMARY.lua')
hifv_testBPdcals()
hifv_semiFinalBPdcals()
```

[4] https://github.com/MoonlitMoo/casa-pipeline/tree/main/hifv
[5] https://almascience.eso.org/processing/documents-and-tools/cycle12/reference-manual-2025
[6] https://casadocs.readthedocs.io/en/stable/index.html

```
hifv_aoflagger(
    flag_target='SECONDARY', aoflagger_file='SECONDARY.lua')
hifv_solint()
hifv_fluxboot()
hifv_aoflagger(
    flag_target='CALIBRATORS', aoflagger_file='CALIBRATORS.lua',
    use_corrected=True)
hifv_finalcals()
hifv_applycals()
hifv_aoflagger(
    flag_target='SCIENCE_TARGET', aoflagger_file='SCIENCE.lua',
    use_corrected=True)
hifv_statwt()
hifv_plotsummary()
hifv_exportdata()
h_save()
```

3.2.1 Initial Flagging

The `hifv_flagdata` task is used to perform deterministic (apriori) flagging. These flagging categories include antenna shadowing, unwanted intents, autocorrelation, edge channels, clipping, quacking, and basebands. Antenna shadowing occurs when the target is close to the horizon, as antennas may block the field of view of antennas behind them. CASA automatically models the positioning of antennas against the source in order to remove shadowed antennas. Unwanted intents are scans in the MS that are not science targets or necessary for calibration. In Listing 3.1, these can be seen to primarily system configuration related or unspecified scans. Auto-correlation is the correlation of the antenna to itself, these zero distance baselines are removed. Edge channel flagging removes the first and last channel of each spectral window, as the interferometer is least sensitive at these boundaries. Clipping removes any visibility amplitudes below zero. Quacking flags a short period at the start and end of each scan. Since the antenna physically moves (slews) between scans, this removes data measured while the antennas may still be settling. Baseband flagging removes ten channels at the start and end of each baseband, analagous to the spectral window flagging. For these observations, there are two basebands covering 8–10 GHz and 10–12 GHz.

3.2.2 Primary calibrator flux scale

The visibilities measured by each antenna have a direction-dependent gain and phase scale. These relative amplitudes and phases must then be corrected in order to measure the correct flux from a

source. Primary calibrators are unresolved point sources with high flux densities (typically >> 10 Jy at λ = 1 m) and stable over long timescales (years). CASA implements the Perley–Butler flux density scale, spanning the frequency range 50 MHz to 50 GHz (Perley and Butler, 2017). This scale contains polynomial flux models of several primary calibrators, allowing the visibilities to be scaled to minimise errors to the model. This assumes an accurate model and low time-variability of the primary calibrator between the modelling and observation. The Perley–Bulter scale quotes accuracy to 3–5% over the full frequency range. CASA recommends observations of 3C286, 3C48, 3C147, or 3C138 to set the absolute flux scale. For the observed clusters, the nearest primary calibrators were 3C286 or 3C48 (see Table 3.2). Scans of the primary calibrators were performed at the start or end of each observing block.

Due to the small number of primary calibrators, the likelihood of a science target lying in close proximity to one is low. This necessitates the use of secondary calibrators, which are also unresolved compact sources but may exhibit significant time variability on the timescale of months to years. The list of VLA calibrators is an inexhaustive list of approximately 1862 usable sources.[7] The absolute flux density scale obtained from the primary calibrator is used to scale the flux density of the secondary sources, as precise emission values are difficult to estimate due to the time-variability. These closer sources then allow for more accurate direction-dependent gain and phase solutions for the science targets.

3.2.3 Prior calibrations

Several deterministic calibrations are monitored and saved during the observations or defined by the instrument setup. Those applied during `hifv_priorcals` include gain curves, atmospheric opacity, antenna position corrections, and requantisation gains. Gain curves account for elevation-dependent gains, where the angle of the antenna causes gravitational deformation of the dish surface. Changes in elevation also alter the atmospheric path-length, which is not entirely transparent to radio emission. The atmospheric opacity correction adjusts for this attenuation, while other atmospheric effects (re-emitted radio emission, delay errors due to density fluctuations) are corrected by later fitted calibrations. The geographical coordinates of antennas may have small time-dependent errors over the course of the observation, which result in delay and baseline errors when correlated. This calibration references an online database of antenna position corrections to apply adjustments. Requantiser gains are amplitude offsets introduced when visibilities are re-quantised to a lower bit size to reduce data rates. These effects are antenna and time-dependent and estimated during the observation.

[7] https://science.nrao.edu/facilities/vla/observing/callist

3.2.4 Calibrator flagging

Before using the primary calibrator observation to fit calibrations, it is flagged for RFI using the custom step `hifv_aoflagger` discussed previously. This step is the first flagging applied to this source and operates on the uncalibrated visibilities. This may share a script and parameter set with other calibrators or science targets, depending on the similarity of the RFI present. These parameters were tested prior to running the full calibration pipeline to optimise the mitigation of RFI signals and preservation of source data. A second round of flagging is performed prior to the `hifv_finalcals` step, described in a following section. The second flagging step operates on the corrected data, which provides the closest representation to the true source. These two passes improve the RFI detection as the quality of the visibilities increase, resulting in cleaner gain and phase solutions during the final calibration stage.

Similarly to the primary calibrators, the secondary calibrators are flagged prior to fitting any calibrations. A first flagging run is performed prior to setting the flux density scale on the raw visibilities. A second pass is performed prior to creating the final calibrators on the corrected visibilities.

3.2.5 Delay and Bandpass calibrations

Errors in antenna position and time measurements cause systematic errors in the measured delay and phases between antennas. These effects are antenna-dependent and can be corrected using observations of the primary calibrator. This assumes one antenna as the reference with zero delay, and calibrates the relative delays of all other antennas. This provides one solution per antenna, per spectral window, and per parallel correlation (LL, RR). To mitigate time-dependent phase errors, a test phase calibration is fitted per integration interval and applied prior to the delay calibration. Following the delay calibration, residual frequency-dependent errors in the phase and gain remain. This is correlated with the spectral window subbands and the basebands, resulting in a total bandpass response. These errors are also corrected using observations of the primary calibrator. To reduce time-dependent amplitude errors, a test complex gain calibration is created over the integration timescales. The test amplitude, test phase, and delay calibrations are then used to construct the bandpass calibration. The step `hifv_testBPdcals` iteratively constructs a self-consistent delay and bandpass calibration for the primary calibrator. The prior, delay, and initial bandpass calibrations are then applied to the primary calibrator visibilities. The following step `hifv_semiFinalBPdcals` recalculates these calibrations and applies them to the primary and secondary calibrator data.

3.2.6 Secondary calibrator flux scale

With the base calibrations completed, the flux density scale can now be transferred from the primary calibrator to the secondary calibrator. In order to accurately transfer a known scale to a source of unknown flux, a well-constrained complex gain calibration is necessary. This uses a default 1 Jy point source model at the phase centre, as a model is not known for the secondary calibrator at this point. To reduce the time-variability in the complex gain of the secondary calibrator, a minimal time solution interval is found using the `hifv_solint` task. The minimum possible interval is the integration timescale, however, a high flagging fraction may make this infeasible. This is then used during the `hifv_fluxboot` task to create a short-timescale phase solution. These corrections are used to fit a complex gain solution with outlier solutions flagged during the fit. This normalised model of the source amplitude is then scaled using the absolute flux density scale of the primary calibrator, and a polynomial model is fit to the resulting spectrum. This scaled model then represents the predicted true spectrum of the secondary calibrator and added to the dataset. Note that any missing spectral windows or antennas in the primary calibrator solutions are propagated to the secondary calibrator and subsequent calibrations.

3.2.7 Final calibration

With flux density models established for the calibrators and flagging performed on the corrected visibilities, the final calibrations can be fitted using the `hifv_finalcals` task. This step first refits the delay and bandpass calibrations using the primary calibrator. An average phase solution is then computed over all scans to normalise the time-dependent phase errors. The short-timescale phase corrections are then computed for all calibrators and used to derive the final complex gain and phase solutions. The final calibration list then consists of the prior, delay, bandpass, average phase, complex gain, and phase calibrations. These calibrations are then applied to obtain the final corrected visibilities. This includes applying the calibrations from the secondary calibrators to the science targets. These solutions are more accurate than those derived from the primary calibration, as the smaller angular seperation reduces interpolation errors in the gain and phase. Once again, any missing spectral windows or antennas in the secondary calibrator solutions are propagated to the science targets to maintain consistency.

3.2.8 Science target processing

The science targets then undergo a single flagging run performed on the corrected visibilities. Ideally, this would occur without the model subtraction, but this was not implemented in the pipeline step. The default model column of a Measurement Set is a 1 mJy point source located at the phase centre. Subtracting this is equivalent to offsetting the visibilities by a constant, to

which RMS-based detection is insensitive to. As the science targets are primarily comprised of faint and diffuse emission, the visibilities are visually similar to noise, and fluctuations can be attributed to RFI. The benefits gained in using the corrected data, rather than the raw visibilities, for flagging were considered to outweigh the risk of erroneous flagging. The variance of the unflagged visibilities is then weighted per baseline based on the resulting exposure time and the number of unflagged points. This improves the sensitivity of the observations by down-weighting baselines that have less data and therefore a lower signal-to-noise ratio.

3.2.9 Pipeline outputs

The `hifv_plotsummary` task produces several plots to check the quality of the final calibration. This includes phase-vs-frequency and amplitude-vs-frequency of the raw and calibrated visibilities for each calibrator and science target. These, alongside reports produced by the previous pipeline tasks, are collected into a weblog and exported alongside the final calibrated datasets as a single product by `hifv_exportdata`.

The calibrated science fields were split from the full dataset, trimming the polarisation products to the parallel correlations and averaging over frequency and time bins. Averaging over frequency alters the uv distribution since $\boldsymbol{u} = \boldsymbol{b}/\lambda$, which results in radial smearing. A typical maximum limit for frequency bin size $\Delta\nu$ can be found in terms of the antenna diameter D and the maximum baseline B, observing frequency ν, and sensitivity x (in the range 1–4, where lower values are stricter limits). For high frequency VLA data, the sensitivity was set to 4.

$$\Delta\nu = x\frac{\nu D}{4B} \tag{3.1}$$

For VLA dishes of diameter 25 m, maximum baselines $\sim 1.25\times10^5$ m, and a minimum observing frequency of 8 GHz, this gives $\Delta\nu \approx 57$ MHz. For channels of width 2 MHz, this corresponds channel averaging by a factor of 28.5. Erring on the side of caution, frequency averaging was decided over 16 channels. Time averaging can also cause smearing due to the Earth's rotation, with an analogous time bin limit Δt in terms of antenna diameter D, maximum baseline B, sensitivity x (in the range 1–3.7), and the angular frequency of Earth's rotation ω. The maximum value for the sensitivity $x = 3.7$ was chosen to match the spectral averaging.

$$\Delta t = x\frac{D}{4\omega B} \tag{3.2}$$

With $\omega \approx 7.27\times10^{-5}$ rad s^{-1}, the maximal time bin is $\Delta t \approx 91$ s. The time averaging was chosen at 60s based upon this limit. Since these values are independent of a specific observation, the same averaging was applied to all datasets.

Ref eqs (slides somewhere?)

3.3 Calibration exceptions

Two observing runs, ID `25A-157.sb47856643.eb48084985.60734.6548690162` and `25A-157.sb47896788.eb48078564.60733.06076180555`, required manual intervention during calibration. This was due to noisy and inconsistent amplitude and phase solutions for a secondary calibrator, in both datasets, affecting spectral windows 22–31. This resulted in an antenna-dependent spread of amplitudes and phases over these spectral windows following the final calibration. This was caused by the pipeline selecting a long solution interval despite the presence of short-timescale phase variations. This was likely due to high RFI in these spectral windows inflating the minimum timescale estimated during `hifv_solint`. To resolve this, a short-timescale phase calibration was fitted while excluding spectral windows contaminated with RFI to produce a single time-dependent solution per antenna, per correlation. A residual phase and amplitude gradient across the spectral windows remained following this correction. A linear bandpass solution was iteratively fit to this gradient, excluding outlying points until convergence. This phase and bandpass solution was then applied in place of the final complex gain and phase solutions to the affected secondary calibrator and science field. This solution was sufficient to correct the phase issues in both datasets.

The observation `25A-157.sb47856643.eb48084985.60734.6548690162` also contained a mislabelled secondary calibrator scan. This scan was labelled with the intent for bandpass calibration, which caused it to be handled alongside the primary calibrator data. This was fixed using the CASA `defintent` task to set the intent to the correct values.

3.4 Imaging

The collated calibrated datasets were then imaged using CASA's `tclean` task. This implements the CLEAN algorithm discussed in Section 2.5. There are several important imaging parameters that are independent of the source properties, but depend on the instrument configuration and scientific objective. For the calibrated observations, test imaging revealed a synthesised elliptical beam with a full width at half maximum (FWHM), defined as the beam width at half the peak sensitivity, of approximately 2.7×2.2 arcsec. To achieve sufficient spatial sampling, 3–5 pixels across the beam are recommended. Accordingly, a cell size of 0.5×0.5 arcsec was chosen for all imaging. The total image size was set to $2^8 \times 3^2 = 2304$ pixels per side to achieve an angular diameter of 19.2×19.2 arcminutes.[8] To prevent radial clipping due to the primary beam truncation, the fractional limit `pblimit` was set to -1×10^{-6}. This angular size captures the full minihalo region for all clusters and was sufficiently wide to image any nearby sources causing PSF artefacts in the central region.

[8] Imaging is most efficient when the image size is even and factorisable into combinations of 2, 3, and 5 only.

These observations require continuum (8–12 GHz) imaging at several angular scales due to the relative sizes of AGN and minihalos. As such, the spectral definition mode set to multi-frequency synthesis `mfs`, the deconvolder (minor cycle algorithm) to multi-term (i.e. multi-scale when provided scales) multi-frequency synthesis `mtmfs` (Rau and Cornwell, 2011), and the `gridder` was left as standard due to negligible widefield effects. As the total intensity is of interest, Stokes I imaging was used, which averages the two parallel polarisations $I = \frac{RR+LL}{2}$ for an intensity measurement. The gain, or fraction of source flux subtracted from the residual during CLEAN cycles, was left at the default of 0.1. These parameters, summarised in Table 3.3, define the general spatial and spectral sampling and deconvolution methods necessary for imaging minihalos.

Table 3.3: General imaging parameters for `tclean` used for all imaging.

Parameter	**Value**
`imsize`	2304×2304 pixels
`cell`	0.5×0.5 arcsec
`pblimit`	-1×10^{-6}
`specmode`	`mfs`
`deconvolver`	`mtmfs`
`gridder`	standard
`stokes`	I
`gain`	0.1

The imaging can be further tuned by changing the `weighting` of visibilities. Natural weighting applies the baseline visibility weights determined during the calibration `statwt` task. This results in a weighting grid across the uv-plane that follows the sampling density. This maximises the beam sensitivity, but negatively impacts the PSF. This can result in brighter and more widely spread sidelobes convolved into the image, influencing the cleaning process. In contrast, uniform weighting normalises the base weights by the total weight of all data at each uv coordinate. This results in a close to uniform weighting across the uv plane which suppresses PSF sidelobes at the cost of decreased sensitivity. Briggs weighting creates smoothly varying scale between the natural and uniform schemes using a `robust` parameter r. For $r = -2$ briggs weighting matches uniform, while $r = 2$ corresponds to natural weighting. This flexibility meant that briggs weighting was chosen, and the robust parameter was altered depending on the intention of the image. Positive robust values of 0.5 and 1 were used to image the minihalo by maximising the synthesised beam sensitivity. Negative robusts of -0.5 and -1 were used when imaging AGN.

To model emission at different angular scales with the `mtmfs` deconvolver, a list of CLEAN scales can be provided. These scales correspond to possible pixel diameters of Gaussian components added to the model. This allows a more complex image to be modelled and is especially important to model the extended emission of minihalos. The scales used during imaging were

determined based upon the observed scale of emissions, but in general corresponded to zero (point source) and powers of two, [0, 2, 4, 8...]. Point sources are a recommended scale to use, with the next smallest corresponding to the beam size, which is approximately four pixels for these images. Intermediate scales were used during exploratory imaging for some clusters when improvements in model quality were observed. The selection of different scales can also be adjusted using the `smallscalebias`, which biases the scale selection towards larger (-1) or smaller scales (1). This was occasionally used to assist in modelling compact sources and set to 0.5.

The final cleaning parameter is the `threshold`, which defines the minimum residual level to clean to. This was altered depending on the RMS σ of the residual image and the lower limit was set in the range 3–5σ for automated cleaning ($\sim$ 10 μJy), and close to σ ($\sim$ 3 μJy) for manual cleaning. For limits close to σ, spurious noise peaks may be incorporated into the model which can negatively impact the resulting image. To prevent this masking is usually used to define regions of true emission that are then cleaned.

Masking can be performed manually and interactively, however, automasking is implemented for `tclean`. This has several thresholds (with CASA default), including `noisethreshold` (5) to prevent masking emission near the current residual RMS, `sidelobethreshold` (1.25) to prevent adding bright sidelobes into the mask, `negativethreshold` (0) to mask negative features, and `lownoisethreshold` (2) to grow the current mask border using binary dilation. Automatic masking was generally well-behaved with these parameters, however, the `sidelobethreshold` was increased to values of 2.5 and 5 for some clusters that featured a PSF with bright and frequent sidelobes.

3.4.1 Self calibration

With this understanding of imaging, the angular resolution and noise levels are then limited by residual antenna-based phase and gain errors. These arise from the interpolation from solutions at the secondary calibrator sky position to the science target position. The brightness model created during the cleaning can be used to fit solutions and improve the calibration model. This is called self calibration as it is an iterative and self-consistent calibration method. Several rounds of phase-only calibration is performed for each cluster, which improves the spatial resolution of the emission.

Self calibration was performed on the joint image produced by all datasets for a cluster. This combines higher angular resolution modelling of compact sources from the C-configuration data, with the more sensitive flux detection of D-configuration visibilities. The combined model is then used to fit phase solutions over the scan interval per antenna, per spectral window, and per correlation. This assumes that short time dependencies have already been corrected during the secondary calibrator. This calibrated data is then used for imaging and a new model is created

and the process is repeated until the phase solutions converge. This was done in several stages for each cluster, with initial images using a negative robust $r = -0.5$, small scales $[0, 2]$, and high thresholds 0.1–0.05 mJy to create accurate models of bright and compact sources. Once a desirable model was achieved, the robust was swapped to positive $r = 0.5$, larger scales were added $[0, 2, 4, 8]$, and the threshold was lowered towards 3σ. This process assisted in preventing prematurely deep cleaning, which can create model artefacts such as negative point sources and virtual sidelobe emission. The parameters discussed in the prior section were adjusted on a per-cluster basis in order to obtain physical and accurate models for self calibration. The notes taken and parameters for each cluster are available in the associated repository under `Image-Processing/<cluster_name>/full_image`.

Between self calibration rounds, the datasets were assessed for highly flagged spectral windows, remaining RFI in visibility amplitudes, and inconsistent phase solutions. Spectral windows where less than $\sim 25\%$ of data remained were flagged as they decreased the signal-to-noise ratio. Spectral windows where phase solutions did not converge were flagged as this indicated that solutions were dominated by noise. The visibilities were also clipped to a maximum value to remove remaining spikes, with the threshold in the range 0.1–0.5 mJy based on the observed visibilities.

Time variability of AGN was an issue noticed during calibration of some clusters. Due to the year offset between the C and D-configuration observations, several bright AGN were observed to have significantly different peak fluxes ($\approx$ 20–40%). This resulted in deep negative point sources around these AGN as the two flux levels were inconsistent within a single brightness model. Two methods were used to correct for this, individual modelling or flux scaling, depending on the improvement in imaging obtained. The first method modelled each dataset individually alongside the joint image. The individual model of the variable source was then spliced into the joint image and applied to the dataset. This allowed the benefits of the joint image to persist for the majority of the image, while correcting the model for the variable source. This was best applied to localised sources where the boundaries of the splice did not cause spatial discontinuities in the model. For variable sources embedded in larger scale emission, such as the minihalo, flux scaling was used instead. This also produced an individual model, but fit the peak brightness of the variable source. This was then used to scale the emission of the joint image in a region around the variable source. This mitigated large discontinuities in the model, but preserved any existing negative pixels. This is noted during the results section for clusters that required these methods.

Chapter 4

How to Analysis

This chapter starts with an overview on imaging and how that is performed. Then we talk a bit about how we actually want to measure the flux of the minihalos. Could combine with actual results.

4.1 Final datasets

- Could have a table here talking about each minihalo, how much data is remaining, spws that are flagged out, some sort of visual.
- Maybe bar graph of spws, line or point graph for minihalo C/D (either total or differentiated by style) normalised to [0, 1]

4.2 Calibration Error

- Calibration error exists, and sources can vary in their flux amount.
- We can try and constrain the error and variance by comparing the secondary calibrator models for D config and the point source fluxes for the images.
- Fluxes for secondaries can vary a lot over a year (100%?) maybe, but lower per month. Can then compare to differences in sources in the image.
- If a trend is observed, likely calibration error. If no trend or close models, constrain calibration error and can see the variance in AGN etc.
- Can retrieve the point source flux by cutting out diffuse sources (uvrange) and by masking to a tight border.

4.3 Calculating the minihalo flux

- We can do this in a variety of ways, but we don't want the AGN contamination so we should subtract this out.
- Do this per dataset to account for variation over time, incorrect subtractions can result in negative bowls in emission, or remaining flux.
- Other problematic sources are dealt with the same way.
- Image the subtracted data and then we create a contour for significant emission (3 sigma) from the residual RMS.
- Calculate total flux in the region.
- Then we calculate the error using this formula from G14 modified for the AGN error since we think its lower.

Chapter 5

Results

- Sections for each minihalo with a brief explanation of imaging process, custom things we had to do to get results.
- Number of self calibration, sources subtracted, difficulties in imaging certain aspects.
- Discussion on the calibration error and variance in the observed point sources.
- Includes an image of the uvrange data showing point sources, image of the minihalo with contours, and anything else we desire.
- Includes discussion compared to previous results or measured values (likely G14) and any caveats or concerns.

Chapter 6

Discussion

- General findings from the above
- Any downsides caveats for the whole dataset
- Compare general results to things like G14 / G19 and how comparable these values are. Both literally and if valid comparison.

Chapter 7

Conclusion

- I'm sure I will have some conclusions here.

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Appendix A

Example AOFLAGGER Lua script

The AOFLAGGER strategy used for the flagging of RFI in this thesis consists of two targeted passes, and a final general pass, designed to address the different RFI environments present across the full 8–12 GHz frequency band. An example of the script is provided in this appendix, the complete scripts used during the calibration process are available in the associated GitHub repository under 'Data Processing' and the relevant observation ID.[1]

The options function defines the spectral windows that are flagged by each pass, and the corresponding execution function. The first pass (execute_awr) targets spectral windows 10–11, corresponding to frequencies ≈9.3–9.5 GHz which contain intermittent satellite RFI. The second pass (execute_microwave) covers spectral windows 22–31 (approximately 10.7–12 GHz), where continuous ground-based communication interference is present. The final pass (execute_final) applies to the full frequency band 8–12 GHz. The targeted passes use independent lower flagging thresholds than the final pass, providing adaptability to the amount of RFI measured in an observation. The values shown are representative of those used in the calibration process, with thresholds ranging from 0.65 to 5.0.

Listing A.1: Example AOFLAGGER Lua script.

```
aoflagger.require_min_version("3.0")

function options()
  opts1 = {
    ["bands"] = {10,11},
    ["execute-function"] = "execute_awr",

  }

  opts2 = {
    ["bands"] = {22,23,24,25,26,27,28,29,30,31},
```

[1] https://github.com/MoonlitMoo/Masters-2025

```
    ["execute-function"] = "execute_microwave"
  }

  opts3 = {
    ["execute-function"] = "execute_final"
  }

  return {
    ["1 - weather radars"] = opts1,
    ["2 - microwave link"] = opts2,
    ["3 - final sweep"] = opts3
  }
end

function execute_awr(input)
    -- 9.3-9.5 GHz (airborne weather radars)
    trim_rfi(input, 1.0, 1.0)
end

function execute_microwave(input)
    -- 10.7-12 GHz (Microwave link + geostationary satellite near
       11.7-12 GHz)
    trim_rfi(input, 1.0, 1.0)
end

function execute_final(input)
    -- Run over remaining at the end as cleanup
    trim_rfi(input, 2.0, 2.0)
end

function trim_rfi(input, base_threshold, transient_threshold_factor)
  -- base_threshold, lower means more sensitive detection
  -- transient_threshold_factor, decreasing this value makes detection
     of transient RFI more aggressive
  local flag_polarizations = input:get_polarizations()
  local flag_representations = { "amplitude" }
  local iteration_count = 4 -- how many iterations to perform?
  local threshold_factor_step = 1.75 -- How much to increase the
     sensitivity each iteration?
  local exclude_original_flags = true
  local frequency_resize_factor = 1.0 -- Amount of "extra" smoothing in
     frequency direction

  local inpPolarizations = input:get_polarizations()

```

```
53  if not exclude_original_flags then
54    input:clear_mask()
55  end
56  local copy_of_input = input:copy()
57
58  for ipol, polarization in ipairs(flag_polarizations) do
59    local pol_data = input:convert_to_polarization(polarization)
60    local converted_data
61    local converted_copy
62
63    for _, representation in ipairs(flag_representations) do
64      converted_data = pol_data:convert_to_complex(representation)
65      converted_copy = converted_data:copy()
66
67      for i = 1, iteration_count - 1 do
68        local threshold_factor = threshold_factor_step ^ (
            iteration_count - i)
69
70        local sumthr_level = threshold_factor * base_threshold
71        if exclude_original_flags then
72          aoflagger.sumthreshold_masked(
73            converted_data,
74            converted_copy,
75            sumthr_level,
76            sumthr_level * transient_threshold_factor,
77            true,
78            true
79          )
80        else
81          aoflagger.sumthreshold(converted_data, sumthr_level,
              sumthr_level * transient_threshold_factor, true, true)
82        end
83
84        -- Do timestep & channel flagging
85        local chdata = converted_data:copy()
86        aoflagger.threshold_timestep_rms(converted_data, 2.0)
87        aoflagger.threshold_channel_rms(chdata, 1.5 * threshold_factor,
            true)
88        converted_data:join_mask(chdata)
89
90        -- High pass filtering steps
91        converted_data:set_visibilities(converted_copy)
92        if exclude_original_flags then
93          converted_data:join_mask(converted_copy)
94        end
```

```lua

        local resized_data = aoflagger.downsample(converted_data, 3, frequency_resize_factor, true)
        aoflagger.low_pass_filter(resized_data, 21, 31, 2.5, 5.0)
        aoflagger.upsample(resized_data, converted_data, 3, frequency_resize_factor)

        local tmp = converted_copy - converted_data
        tmp:set_mask(converted_data)
        converted_data = tmp

        aoflagger.visualize(converted_data, "Residual #" .. i, i + iteration_count)
        aoflagger.set_progress((ipol - 1) * iteration_count + i, # flag_polarizations * iteration_count)
      end -- end of iterations

      if exclude_original_flags then
        aoflagger.sumthreshold_masked(
          converted_data,
          converted_copy,
          base_threshold,
          base_threshold * transient_threshold_factor,
          true,
          true
        )
      else
        aoflagger.sumthreshold(converted_data, base_threshold, base_threshold * transient_threshold_factor, true, true)
      end
    end -- end of complex representation iteration

    if exclude_original_flags then
      converted_data:join_mask(converted_copy)
    end

    -- Helper function used below
    function contains(arr, val)
      for _, v in ipairs(arr) do
        if v == val then
          return true
        end
      end
      return false
    end
```

```
135
136     if contains(inpPolarizations, polarization) then
137       if input:is_complex() then
138         converted_data = converted_data:convert_to_complex("complex")
139       end
140       input:set_polarization_data(polarization, converted_data)
141     else
142       input:join_mask(converted_data)
143     end
144
145     aoflagger.visualize(converted_data, "Residual #" .. iteration_count,
            2 * iteration_count)
146     aoflagger.set_progress(ipol, #flag_polarizations)
147   end -- end of polarization iterations
148
149   if exclude_original_flags then
150     aoflagger.scale_invariant_rank_operator_masked(input, copy_of_input,
            0.2, 0.2)
151   else
152     aoflagger.scale_invariant_rank_operator(input, 0.2, 0.2)
153   end
154
155   aoflagger.threshold_timestep_rms(input, 4.0)
156
157   input:flag_nans()
158 end
```